Golden Prime Symmetry

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Abstract

The new theorems of **Golden Prime Symmetry** proves geometrically the existence of an equivalence relation on the infinite set of prime numbers. The golden ratio it induces a partition into 8 equivalence classes corresponding to 8 rotation invariant angles or isometries of 2 opposite pentagons rotated in the complex plane. Moreover, when arranged in matrices they form a Lie rotation group. The quotient set of residual prime classes corresponds to the zeros of a minimal irreducible polynomial over $\mathbb{Q}$ corresponding to a cyclotomic polynomial of order 5. Analogously, there is another equivalence relation on the infinite set of the twelfth Fibonacci numbers which has the same properties for two other opposite pentagons rotated in the complex plane. Thus, it is shown that the sine function maps the prime numbers into unique golden images that depend on the last digit of the prime $(1, 3, 7, 9)$.

Key words: *Golden ratio, Prime numbers, Twelfth Fibonacci, Cyclotomic polynomial.*

1 Introduction

To begin, we define the golden ratio, the 16 golden angles in unit circle and Midy's theorem, which are the basis of golden prime symmetry.

1.1 Golden ratio

Definition 3 of Book VI of Euclid's Elements (300 - 265 BC), defines the golden ratio as follows: *"A line is said to have been cut in extreme and half reason when the entire line is to the larger segment as the larger segment is to the smaller segment"*.

Let be $\frac{a}{b} = \frac{a+b}{a} = \frac{a}{a} + \frac{b}{a} = 1 + \frac{1}{a/b}$ if $x = a/b$

$x = 1 + \frac{1}{x} \quad \Rightarrow x^2 = x + 1 \quad \Rightarrow x^2 - x - 1 = 0$

whose zeros are: $x^2 - x - 1 = \left(x - \frac{1+\sqrt{5}}{2}\right)\left(x - \frac{1-\sqrt{5}}{2}\right) = (x - \varphi)(x + \varphi^{-1})$

$x_1 = \frac{1+\sqrt{5}}{2} \qquad x_2 = \frac{1-\sqrt{5}}{2}$

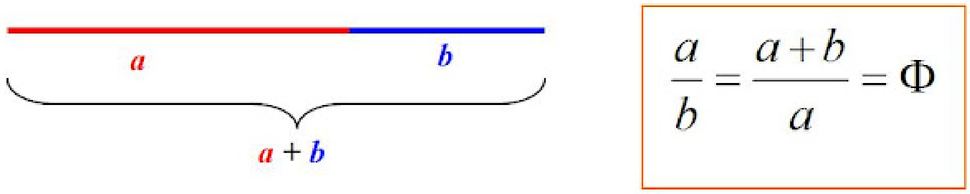

Figure 1: Major and minor segment.

1.2 Fibonacci sequence

The Fibonacci sequence was described in Europe by Leonardo of Pisa and published in his work "Liber Abaci" in 1202. It is defined as a recurrence sequence, where each term is obtained from the sum of the two previous terms.

$F_n = F_{n-1} + F_{n-2}$ Is obtained: $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597...$

https://oeis.org/A000045

The golden ratio is also the limit of the quotient of the nth Fibonacci term and its successor.

$$\lim_{x\to\infty} \frac{F_{n+1}}{F_n} = \lim_{x\to\infty} \frac{F_n + F_{n-1}}{F_n} = \lim_{x\to\infty} 1 + \frac{F_{n-1}}{F_n} = 1 + \lim_{x\to\infty} \frac{F_n}{F_{n-1}} = 1 + \frac{1}{L}$$

$$L = L + \frac{1}{L} \quad \Rightarrow \quad L^2 - L - 1 = 0 \quad \Rightarrow \quad L_1 = \frac{1+\sqrt{5}}{2} \qquad L_2 = \frac{1-\sqrt{5}}{2}$$

1.3 Trigonometry and golden angles

There are 16 angles in the interval $(-2\pi, 2\pi)$ whose sine and cosine image is the golden ratio.

$$\begin{array}{ll}
2\sin(18)^\circ = \varphi^{-1} & 2\cos(36)^\circ = \varphi \\
2\sin(54)^\circ = \varphi & 2\cos(72)^\circ = \varphi^{-1} \\
2\sin(126)^\circ = \varphi & 2\cos(108)^\circ = -\varphi^{-1} \\
2\sin(162)^\circ = \varphi^{-1} & 2\cos(144)^\circ = -\varphi \\
2\sin(198)^\circ = -\varphi^{-1} & 2\cos(216)^\circ = -\varphi \\
2\sin(234)^\circ = -\varphi & 2\cos(252)^\circ = -\varphi^{-1} \\
2\sin(306)^\circ = -\varphi & 2\cos(288)^\circ = \varphi^{-1} \\
2\sin(342)^\circ = -\varphi^{-1} & 2\cos(324)^\circ = \varphi
\end{array}$$

They can also be expressed in the form of quotients

$$\frac{\sin(72)^\circ}{\sin(36)^\circ} = \frac{\sin(108)^\circ}{\sin(144)^\circ} = \frac{\sin(252)^\circ}{\sin(216)^\circ} = \frac{\sin(288)^\circ}{\sin(324)^\circ} = \varphi$$
$$\frac{\cos(18)^\circ}{\cos(54)^\circ} = \frac{\cos(162)^\circ}{\cos(126)^\circ} = \frac{\cos(198)^\circ}{\cos(234)^\circ} = \frac{\cos(342)^\circ}{\cos(306)^\circ} = \varphi$$
$$\frac{\sin(54)^\circ}{\sin(18)^\circ} = \frac{\sin(126)^\circ}{\sin(162)^\circ} = \frac{\sin(234)^\circ}{\sin(198)^\circ} = \frac{\sin(306)^\circ}{\sin(342)^\circ} = \varphi^2$$
$$\frac{\cos(36)^\circ}{\cos(72)^\circ} = \frac{\cos(144)^\circ}{\cos(108)^\circ} = \frac{\cos(216)^\circ}{\cos(252)^\circ} = \frac{\cos(324)^\circ}{\cos(288)^\circ} = \varphi^2$$

1.4 Midy's Theorem

Let a/p be a fraction where $a < p$ and $p > 5$ is a prime number. Suppose further that this fraction has a periodic decimal expansion, where the number of digits in the period is even, i.e.: $\frac{a}{p} = 0.\overline{a_1 a_2 ... a_{2k-1} a_{2k}}$. If we divide the period into two halves or blocks and add them together, we get an integer consisting of only nines.

Proof

If the period P has $2n$ digits, it can be divided into two pieces of n digits that we will call A and B, such that: $P = A.10^n + B$, we have to show that $A + B = 99...99 = 10^n - 1$.

$P = 10^n.A + B = \dfrac{10^{2n} - 1}{p} = \dfrac{(10^n + 1)(10^n - 1)}{p}$

If the left hand side of the equality is an integer, the right hand side must also be an integer. That is, $p|(10^n + 1)$ or $p|(10^n - 1)$. It follows that p cannot divide a number consisting of n nines. The only option is that $(10^n + 1)/p$ is an integer.

$\dfrac{a}{p} = \dfrac{A.10^n + B}{10^{2n} - 1} = \dfrac{10^n.A + B}{(10^n + 1)(10^n - 1)}$, we spend $10^n + 1$ to multiply

$$\frac{a(10^n + 1)}{p} = \frac{10^n.A + B}{10^n - 1} = \frac{10^n.A - A + A + B}{10^n - 1} = \frac{A(10^n - 1) + A + B}{10^n - 1} = A + \frac{A + B}{10^n - 1}$$

Since $p|(10^n + 1$, the left-hand side of the equality is integer, so $(A + B)/(10^n - 1)$ must necessarily be integer, let us designate it k.

$\dfrac{A + B}{10^n - 1} = k \quad \Rightarrow \quad A + B = k(10^n - 1)$

Now, since the initial fraction is periodic, $A + B > 0$ for the period to be nonzero. The largest number that can take an n-digit number consisting of n nines is $10^n - 1$.

Therefore, $A < 10^n - 1$ and $B < k(10^n - 1)$

Then, $A + B < 2(10^n - 1) \Rightarrow \quad 0 < \dfrac{A + B}{10^n - 1} < 2$

In such a case, A is composed of only nines and B is composed of only nines. That is,

$P = 10.A^n + B = 999....999$.

This would indicate that the period is not of even length. The period is of length one and consists of the single digit nine. Then the inequality is strict

$A + B < 2(10^n - 1); \quad A + B \leq (10^n - 1)$

Since the fraction is an integer, the only option is that $k = \dfrac{A + B}{10^n - 1} = 1$

Then, $A + B = 10^n - 1$, that is, if we split the decimal period into two blocks A and B and add them together, the result is a number that is composed of only nines. $\square$.

Example:

$1/7 = 0.\overline{142857}$

$1/11 = 0,\overline{09}$

$1/13 = 0,\overline{076923}$

$1/17 = 0,\overline{0588235294117647}$

$1/19 = 0,\overline{052631578947368421}$

$1/23 = 0,\overline{0434782608695652173913}$

$1/29 = 0,\overline{0344827586206896551724137931}$

$1/31 = 0,\overline{032258064516129}$

Let's look at the chains of nines:

$1/7 : 142 + 857 = 999$
$1/13 : 076 + 923 = 999$
$1/17 : 05882352 + 94117647 = 99999999$
$1/19 : 052631578 + 947368421 = 999999999$
$1/23 : 04347826086 + 95652173913 = 99999999999$
$1/29 : 03448275862068 + 96551724137931 = 99999999999999$
$1/31 : 03225 + 80645 + 16129 = 99999$

2 Theorem 1: Golden Prime Partition

The golden ratio is the sine image of the discrete space of all prime numbers greater than 5*. And the module* 360° *of the prime reciprocal decimal expansion induces a golden partition into* 8 *infinite equivalence classes which are invariant under the isometries of the regular pentagon and its opposite rotation.*

Let be $(10^{p-1} - 1)/p$ *is the decimal expansion of* p *prime, it is satisfied that for all* $p > 5$

$f : \mathbb{P} \to \mathbb{R}$

$p \longmapsto f(p)$

$$f(p) = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = \begin{cases} \varphi & if \quad P_{3odd}, \quad P_{7odd} \\ \varphi^{-1} & if \quad P_{1odd}, \quad P_{9odd} \\ -\varphi & if \quad P_{3even}, \quad P_{7even} \\ -\varphi^{-1} & if \quad P_{1even}, \quad P_{9even} \end{cases}$$

Where $\varphi = \dfrac{1+\sqrt{5}}{2} \qquad \varphi^{-1} = \dfrac{\sqrt{5}-1}{2}$

Moreover, the isometries correspond to 8 *angles which are the residual classes module* 360° *which partition the primes into disjoint families according to their last digit:* P_1, P_3, P_7, P_9*. In turn, each family is further subdivided into 2 subfamilies: those where each penultimate digit is either odd or even as follows:* $P_{1odd}, P_{3odd}, P_{7odd}, P_{9odd}, P_{1even}, P_{3even}, P_{7even}, P_{9even}$.

2.1 Penult-odd Primes

Definition: They are those primes whose penultimate digit is an odd number. The periodic decimal expansion of these primes has 4 residual classes modulo 360° which correspond to the angles of a regular pentagon located in the plane with center at the origin: $18°, 54°, 126°, 162°$. These are congruent primes $(11, 13, 17, 19)$ module 20.

Defined by comprehension:

$P_{1odd} = \{p \in \mathbb{P} / \quad 2p^{-1}(10^{p-1} - 1) \equiv 18° \pmod{360°} \Leftrightarrow p \equiv 11 \pmod{20}\}$
$P_{3odd} = \{p \in \mathbb{P} / \quad 2p^{-1}(10^{p-1} - 1) \equiv 126° \pmod{360°} \Leftrightarrow p \equiv 13 \pmod{20}\}$
$P_{7odd} = \{p \in \mathbb{P} / \quad 2p^{-1}(10^{p-1} - 1) \equiv 54° \pmod{360°} \Leftrightarrow p \equiv 17 \pmod{20}\}$
$P_{9odd} = \{p \in \mathbb{P} / \quad 2p^{-1}(10^{p-1} - 1) \equiv 162° \pmod{360°} \Leftrightarrow p \equiv 19 \pmod{20}\}$

Defined by extension:

$P_{1odd} \Leftrightarrow 18°$ https://oeis.org/A141884

11, 31, 71, 131, 151, 191, 211, 251, 271, 311, 331, 431, 491, 571, 631, 691, 751, 811, 911, 971, ...

$P_{3odd} \Leftrightarrow 126^\circ$ https://oeis.org/A141885

$13, 53, 73, 113, 173, 193, 233, 293, 313, 353, 373, 433, 593, 613, 653, 673, 733, 773, 853, 953, ...$

$P_{7odd} \Leftrightarrow 54^\circ$ https://oeis.org/A141886

$17, 37, 97, 137, 157, 197, 257, 277, 317, 337, 397, 457, 557, 577, 617, 677, 757, 797, 857, 877, 937, ...$

$P_{9odd} \Leftrightarrow 162^\circ$ https://oeis.org/A141887

$19, 59, 79, 139, 179, 199, 239, 359, 379, 419, 439, 479, 499, 599, 619, 659, 719, 739, 839, 859, 919, ...$

2.2 Penult-even Primes

Definition: They are those primes whose penultimate digit is an even number. The periodic decimal expansion of these primes has 4 residual classes modulo 360° that correspond to the angles of a regular inverted pentagon located in the plane with center at the origin: $198^\circ, 234^\circ, 306^\circ, 342^\circ$. These are congruent primes $(1, 3, 7, 9)$ module 20.

Defined by comprehension:

$P_{1even} = \{p \in \mathbb{P}/ \quad 2p^{-1}(10^{p-1} - 1) \equiv 198^\circ \pmod{360^\circ} \Leftrightarrow p \equiv 1 \pmod{20}\}$

$P_{3even} = \{p \in \mathbb{P}/ \quad 2p^{-1}(10^{p-1} - 1) \equiv 306^\circ \pmod{360^\circ} \Leftrightarrow p \equiv 3 \pmod{20}\}$

$P_{7even} = \{p \in \mathbb{P}/ \quad 2p^{-1}(10^{p-1} - 1) \equiv 234^\circ \pmod{360^\circ} \Leftrightarrow p \equiv 7 \pmod{20}\}$

$P_{9even} = \{p \in \mathbb{P}/ \quad 2p^{-1}(10^{p-1} - 1) \equiv 342^\circ \pmod{360^\circ} \Leftrightarrow p \equiv 9 \pmod{20}\}$

Defined by extension:

$P_{1even} \Leftrightarrow 198^\circ$ https://oeis.org/A141881

$41, 61, 101, 181, 241, 281, 401, 421, 461, 521, 541, 601, 641, 661, 701, 761, 821, 881, 941, ...$

$P_{3even} \Leftrightarrow 306^\circ$ https://oeis.org/A105854

$23, 43, 83, 103, 163, 223, 263, 283, 383, 443, 463, 503, 523, 563, 643, 683, 743, 823, 863, 883, 983, ...$

$P_{7even} \Leftrightarrow 234^\circ$ https://oeis.org/A141882

$07, 47, 67, 107, 127, 167, 227, 307, 347, 367, 467, 487, 547, 587, 607, 647, 727, 787, 827, 887, 907, ...$

$P_{9even} \Leftrightarrow 342^\circ$ https://oeis.org/A141883

$29, 89, 109, 149, 229, 269, 349, 389, 409, 449, 509, 569, 709, 769, 809, 829, 929, ...$

In general, it is complied with:

Penult-odd Primes

$2\sin(18)^\circ \quad = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = \varphi^{-1} \qquad \Leftrightarrow p \equiv 11 \pmod{20}; \qquad P_{1odd}$

$2\sin(126)^\circ \quad = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = \varphi \qquad \Leftrightarrow p \equiv 13 \pmod{20}; \qquad P_{3odd}$

$2\sin(54)^\circ \quad = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = \varphi \qquad \Leftrightarrow p \equiv 17 \pmod{20}; \qquad P_{7odd}$

$2\sin(162)^\circ \quad = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = \varphi^{-1} \qquad \Leftrightarrow p \equiv 19 \pmod{20}; \qquad P_{9odd}$

Penult-even Primes

$2\sin(198)^\circ \quad = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = -\varphi^{-1} \qquad \Leftrightarrow p \equiv 1 \pmod{20}; \qquad P_{1even}$

$2\sin(306)^\circ \quad = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = -\varphi \qquad \Leftrightarrow p \equiv 3 \pmod{20}; \qquad P_{3even}$

$2\sin(234)^\circ \quad = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = -\varphi \qquad \Leftrightarrow p \equiv 7 \pmod{20}; \qquad P_{7even}$

$2\sin(342)^\circ \quad = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = -\varphi^{-1} \qquad \Leftrightarrow p \equiv 9 \pmod{20}; \qquad P_{9even}$

Proof

Let θ be an angle of the form $9k$. Now let's obtain the sine of the angle double of θ, which as we have already seen when it is an even multiple of 9 is a golden angle.

Let $\theta = 9k$ with $k \in \mathbb{Z}$

$\sin(2(\theta))^\circ = \sin(2(9k))^\circ = \sin(18k)^\circ$

Midy's theorem guarantees that the reciprocal decimal expansion of prime numbers always add up to strings of nines. In other words, they are $k \equiv 0 \pmod 9$.

Now, let be $\theta = p^{-1}(10^{p-1} - 1) = 9k$

Then, $2\theta = 2(p^{-1}(10^{p-1} - 1)) = 18k$

$2\sin(2\theta)^\circ = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ = 2\sin(18k)^\circ$

Which has only the golden ratio as its image set for all $p \in \mathbb{P}$. $\square$

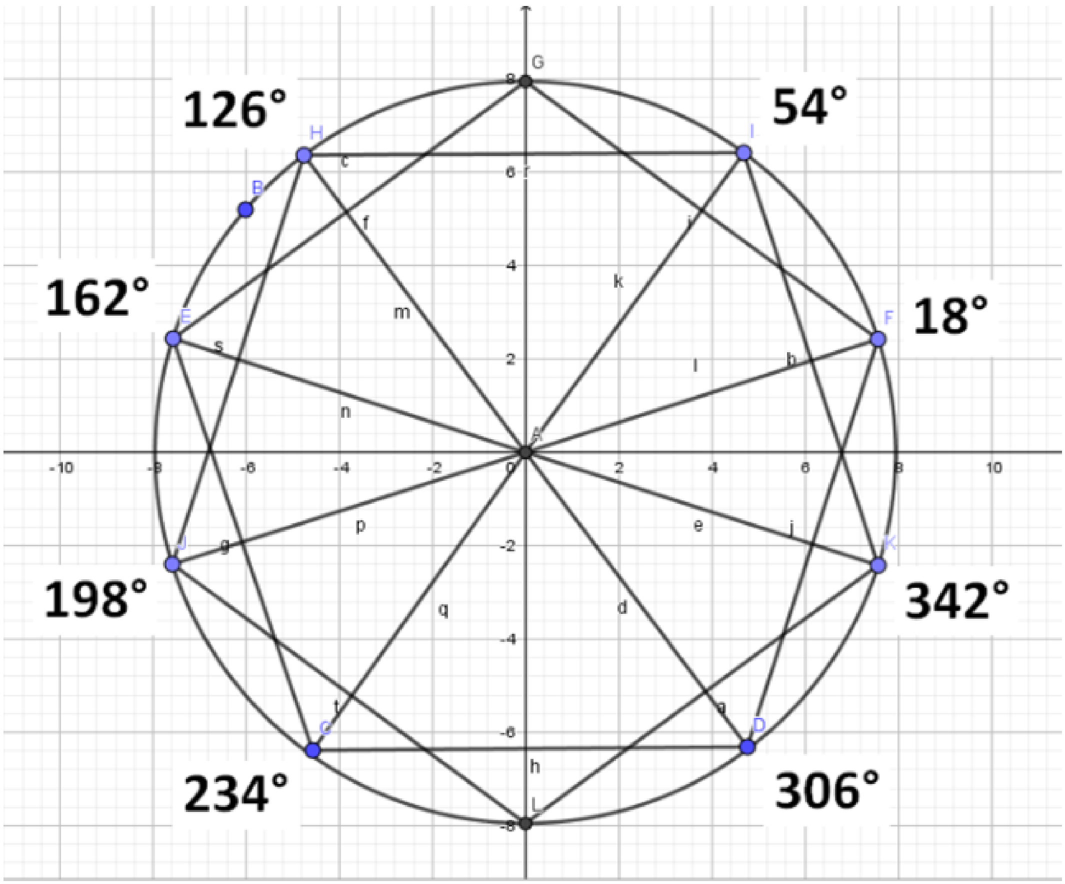

Figure 2: Prime classes at the vertices of two opposite pentagons.

$$\text{Table}\left[\frac{2\left(10^{p_n-1}-1\right)}{p_n} \bmod 360, \{n, 4, 400, 1\}\right]$$

p_n is the n^{th} prime number

Result

{234, 18, 126, 54, 162, 306, 342, 18, 54, 198, 306, 234, 126, 162, 198, 234, 18,
126, 162, 306, 342, 54, 198, 306, 234, 342, 126, 234, 18, 54, 162, 342, 18,
54, 306, 234, 126, 162, 198, 18, 126, 54, 162, 18, 306, 234, 342, 126, 162,
198, 18, 54, 306, 342, 18, 54, 198, 306, 126, 234, 18, 126, 54, 18, 54, 234,
342, 126, 162, 234, 126, 162, 306, 342, 54, 198, 342, 162, 198, 18, 126,
162, 306, 342, 54, 198, 306, 234, 162, 234, 18, 162, 306, 342, 198, 306,
198, 234, 54, 306, 342, 18, 54, 234, 126, 162, 198, 234, 126, 54, 162, 18,
198, 306, 234, 126, 162, 198, 126, 54, 306, 18, 198, 342, 162, 234, 126,
162, 306, 18, 54, 198, 342, 126, 234, 54, 342, 18, 198, 306, 234, 342, 162,
126, 54, 162, 306, 54, 198, 306, 234, 234, 18, 162, 342, 54, 198, 234, 126,
234, 18, 54, 306, 18, 54, 342, 126, 162, 198, 18, 126, 162, 342, 18, 198,
306, 342, 234, 18, 126, 54, 306, 342, 54, 306, 342, 18, 126, 306, 18, 198,
234, 126, 198, 126, 54, 306, 342, 18, 54, 342, 162, 54, 162, 306, 342, 18,
54, 198, 306, 234, 162, 198, 234, 198, 234, 126, 198, 162, 342, 306, 234,
342, 126, 162, 234, 18, 126, 162, 18, 198, 306, 234, 342, 126, 162, 18, 306,
18, 306, 342, 126, 162, 234, 18, 162, 306, 54, 198, 234, 342, 126, 162, 198,
234, 54, 54, 306, 234, 342, 126, 54, 162, 342, 198, 306, 126, 198, 234, 126,
162, 54, 306, 234, 342, 198, 18, 306, 18, 234, 198, 234, 18, 126, 54, 162,
342, 198, 234, 126, 18, 126, 342, 18, 126, 162, 234, 126, 54, 162, 306, 18,
54, 234, 342, 162, 126, 306, 342, 198, 306, 234, 342, 162, 18, 126, 342,
18, 54, 198, 306, 126, 198, 162, 306, 234, 126, 198, 54, 162, 306, 18, 234,
342, 126, 198, 234, 126, 54, 342, 18, 126, 162, 198, 234, 18, 54, 18, 54,
198, 306, 342, 126, 162, 18, 54, 306, 54, 198, 234, 162, 234, 126, 54, 306,
198, 18, 162, 306, 342, 18, 54, 162, 18, 126, 342, 54, 198, 126, 234, 54,

Figure 3: Table of prime classes module 360.

https://www.wolframalpha.com/input?i=Table+%5B+2%28prime%28n%29%29%5E%28-1%29%2810%5E%28prime%28n%29-1%29+-1%29+mod+360+%2C+%7B+n%2C+10000%2C+10300%2C+1+%7D+%5D&lang=es

https://www.wolframalpha.com/input?i=sin%5B+2%28%28104659%29%29%5E%28-1%29%2810%5E%28%28104659%29-1%29+-1%29+%5D%C2%B0++%3D++sin%5B+162+%5D%C2%B0&lang=es

Input

$$\text{Table}\left[2\sin\left(\left(\frac{2\left(10^{p_n-1}-1\right)}{p_n}\right)^{\circ}\right), \{n, 4, 200, 1\}\right]$$

p_n is the n^{th} prime number

Result

$$\left\{\frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}),\right.$$
$$\frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(1-\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(1-\sqrt{5}),$$
$$\frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1-\sqrt{5}),$$
$$\frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}),$$
$$\frac{1}{2}(1-\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(1-\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(1-\sqrt{5}),$$
$$\frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}),$$
$$\frac{1}{2}(1-\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}),$$
$$\frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1-\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}),$$
$$\frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(1-\sqrt{5}),$$
$$\frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1-\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}),$$
$$\frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(1-\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(1-\sqrt{5}),$$
$$\frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}),$$
$$\frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1+\sqrt{5}), \frac{1}{2}(1+\sqrt{5}), \frac{1}{2}(-1-\sqrt{5}), \frac{1}{2}(1-\sqrt{5}), \frac{1}{2}(1+\sqrt{5}),$$

Figure 4: Table of golden prime function.

https://www.wolframalpha.com/input?i=Table+%5B+2+sin%282%28prime%28n%29%29%5E%28-1%29%2810%5E%28prime%28n%29-1%29+-1%29%29%C2%B0%2C+%7B+n%2C+10000%2C+10300%2C+1+%7D+%5D

2.3 Golden Prime Equivalence Relation

There exists an equivalence relation $\sim$ *on the set of prime numbers* $p > 5$ *such that it determines a partition into* 8 *residual classes modulo* 360

The 8 *prime golden classes are:*

$P_{1odd} \equiv 18 \pmod{360}$ $\quad$ $P_{1even} \equiv 198 \pmod{360}$

$P_{3odd} \equiv 126 \pmod{360}$ $\quad$ $P_{3even} \equiv 306 \pmod{360}$

$P_{7odd} \equiv 54 \pmod{360}$ $\quad$ $P_{7even} \equiv 234 \pmod{360}$

$P_{9odd} \equiv 162 \pmod{360}$ $\quad$ $P_{9even} \equiv 342 \pmod{360}$

The 8 residual prime classes are golden angles. Since the angles are invariant for each class, it is satisfied that the 8 classes are disjoint and their union is equal to the entire infinite set of prime numbers $p > 5$. i.e. for all i, j: $\quad P_i \cap P_j = \emptyset$ and $\bigcup P_i = \mathbb{P}$

Quotient set

$\mathbb{P}/\sim = \{[18^\circ], [54^\circ], [126^\circ], [162^\circ], [198^\circ], [234^\circ], [306^\circ], [342^\circ]\}$

Now let's look at the 3 properties of every equivalence relation:

Reflexive: $\forall x \in \mathbb{P} : x \sim x$

i.e. Let be $13 \in P_{3impar}$ y $13 \sim 13$

Simetric: $\forall x, y \in \mathbb{P} : x \sim y \Rightarrow y \sim x$

i.e. Let be $13 \in P_{3impar}$, $\quad 953 \in P_{3impar}$: $\quad 13 \sim 953 \Rightarrow 953 \sim 13$

Transitive: $\forall x, y, z \in \mathbb{P} : x \sim y \wedge y \sim z \Rightarrow x \sim z$

i.e. Let be $13 \in P_{3impar}$, $\quad 953 \in P_{3impar}$, $\quad 1993 \in P_{3impar}$:

$13 \sim 953 \wedge 953 \sim 1993 \Rightarrow 13 \sim 1993$ $\square$

2.4 Golden Prime Classes Penult-odd

Let's see some examples of the penult-odd equivalence classes and their class representatives which are the invariant golden angles: $18°, 54°, 126°, 162°$

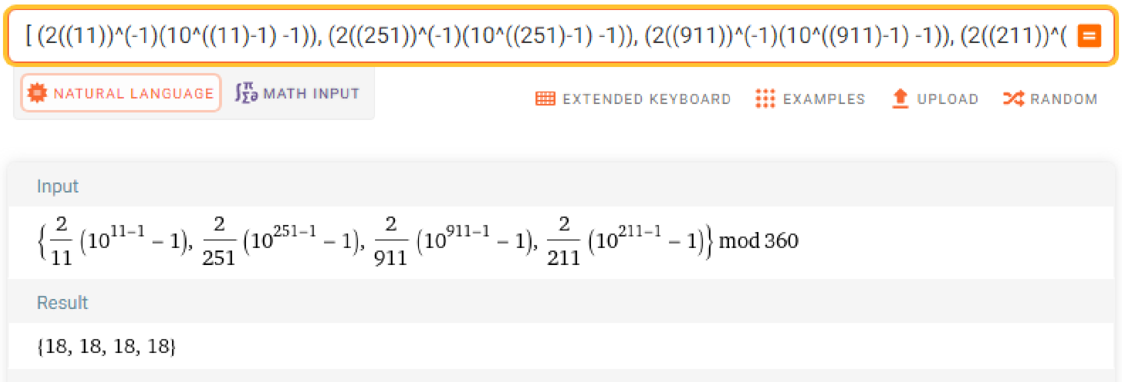

Figure 5: Table of prime class P_{1odd}.

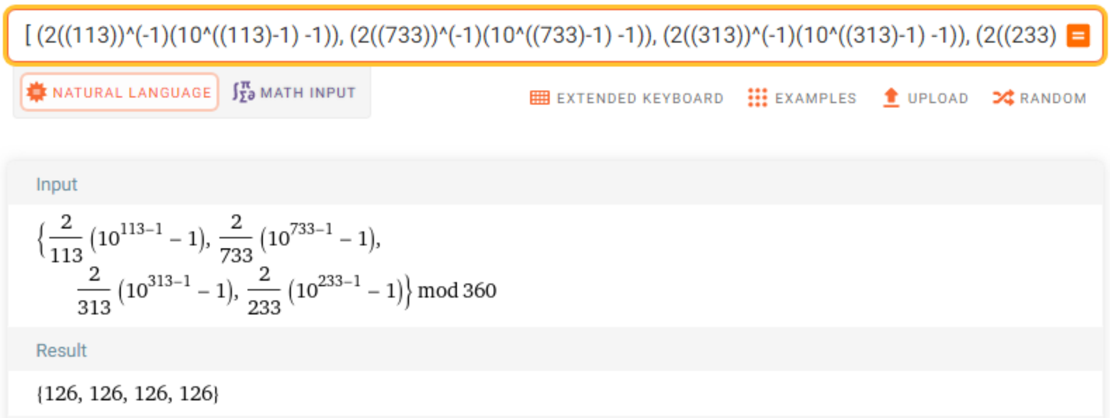

Figure 6: Table of prime class P_{3odd}.

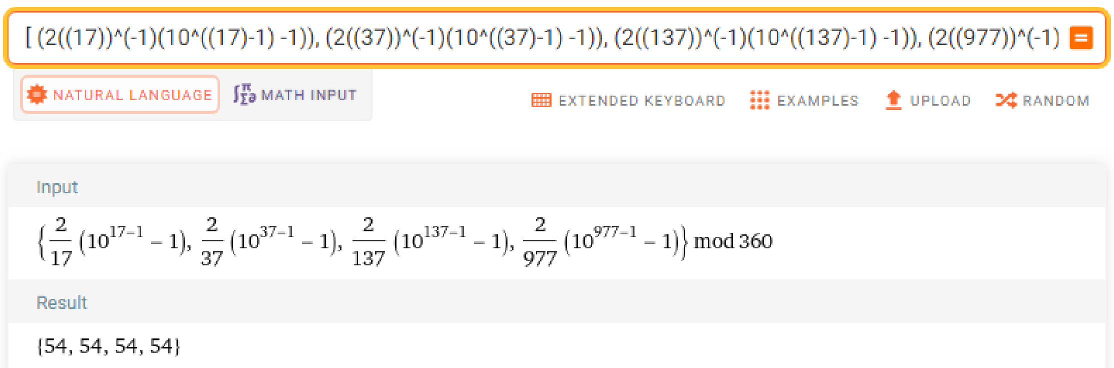

Figure 7: Table of prime class P_{7odd}.

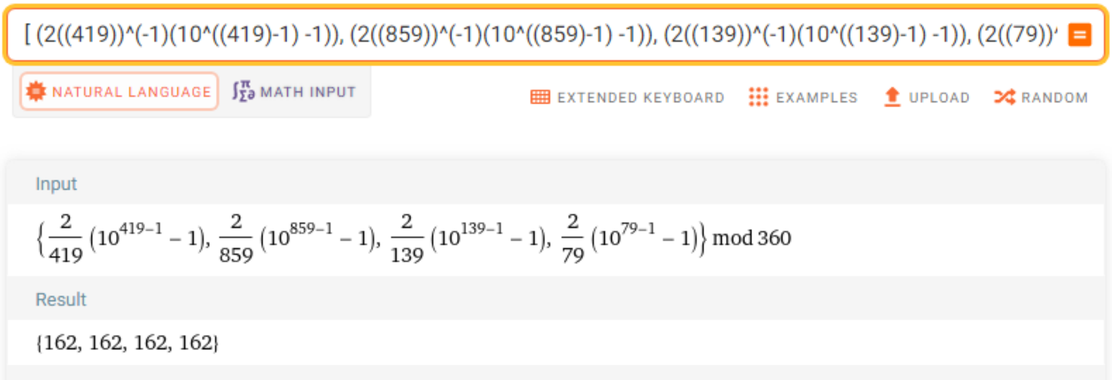

Figure 8: Table of prime class P_{9odd}.

2.5 Golden Prime Classes Penult-even

Let's see some examples of the penulti-pair equivalence classes and their class representatives which are the invariant golden angles: $198°, 234°, 306°, 342°$

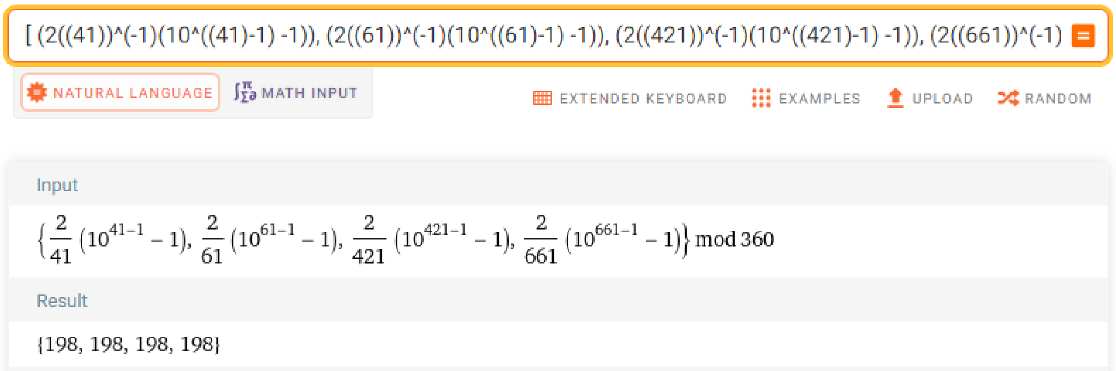

Figure 9: Table of prime class P_{1even}.

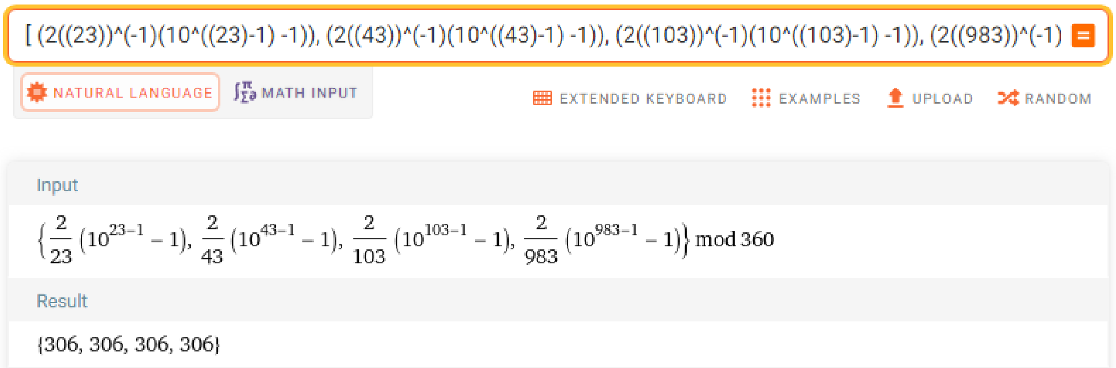

Figure 10: Table of prime class P_{3even}.

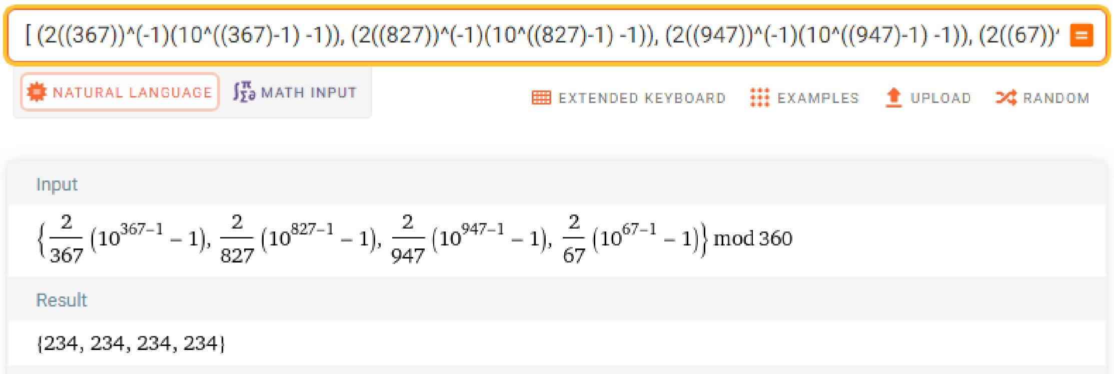

Figure 11: Table of prime class P_{7even}.

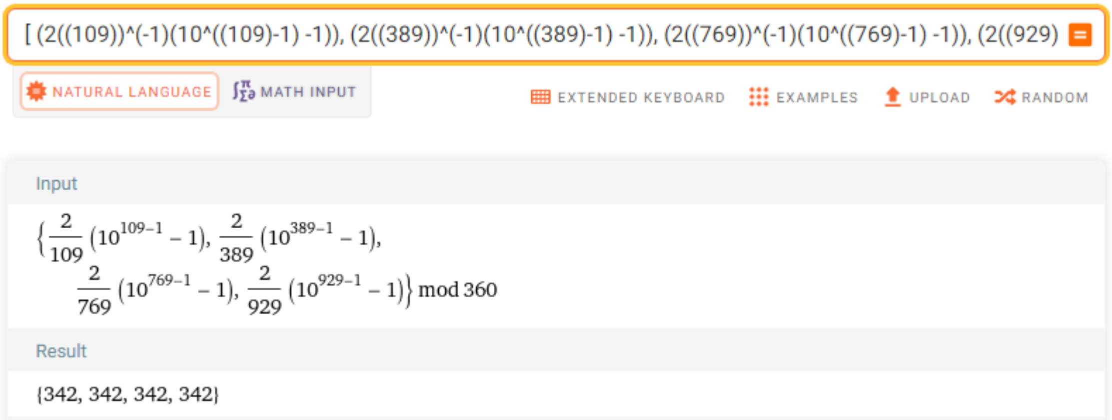

Figure 12: Table of prime class P_{9even}.

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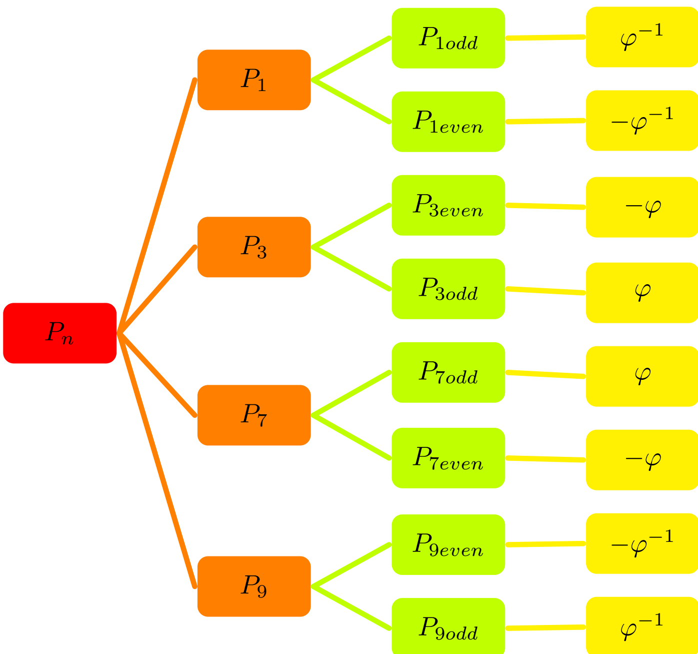

Figure 13: Golden prime tree.

3 Theorem 2: Golden Prime Rotation Matrix

An orthogonal matrix A is a square matrix whose inverse matrix is equal to its transpose matrix. The set of orthogonal matrices is a set of isometric (invariant) transformations of real vector spaces or real Hilbert spaces. This set is the representation of the orthogonal group $O(n)$. Each rotation is described by an orthogonal matrix nxn with real entries, such that doing the product with its transpose yields the identity matrix and with determinant 1. The real group $SO(n)$ is a Lie group and is a subgroup of the orthogonal group $O(n)$. This subgroup $SO(n)$ can be identified with the group of rotations of the space $\mathbb{R}^n$. Let's see the case for a rotation r in $\mathbb{R}^2$.

The golden prime rotation matrix it is a special orthogonal matrix called Toeplitz matrix, which is a square matrix in which the elements of its diagonals from left to right are constant.

i.e. $\forall a_{i,j} \in A \rightarrow a_{i,j} = a_{i+1,j+1}$

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \qquad det(A) = \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = 1$$

The angles $\theta = \{18°, 54°, 126°, 162°, 198°, 234°, 306°, 342°\}$
are the 8 invariant residual prime classes

$$A = \begin{pmatrix} \cos(2p^{-1}(10^{p-1}-1))° & -\sin(2p^{-1}(10^{p-1}-1))° \\ \sin(2p^{-1}(10^{p-1}-1))° & \cos(2p^{-1}(10^{p-1}-1))° \end{pmatrix}$$

$$det(A) = \begin{vmatrix} \cos(2p^{-1}(10^{p-1}-1))° & -\sin(2p^{-1}(10^{p-1}-1))° \\ \sin(2p^{-1}(10^{p-1}-1))° & \cos(2p^{-1}(10^{p-1}-1))° \end{vmatrix} = 1$$

3.1 Eigenvalues of the Golden Prime Matrix

- **For classes P_{1odd} and P_{9even}**

  Characteristic polynomial: $\lambda^2 - \sqrt{2+\varphi}\lambda + 1 = 0$

  $\lambda_1 = \dfrac{\sqrt{2+\varphi}}{2} + \dfrac{\varphi^{-1}}{2}i = e^{i18} \qquad \lambda_2 = \dfrac{\sqrt{2+\varphi}}{2} - \dfrac{\varphi^{-1}}{2}i = e^{i342}$

  Such that: $\lambda_1.\lambda_2 = e^{i18}.e^{i342} = 1$

- **For classes P_{1even} and P_{9odd}**

  Characteristic polynomial: $\lambda^2 + \sqrt{2+\varphi}\lambda + 1 = 0$

  $\lambda_1 = -\dfrac{\sqrt{2+\varphi}}{2} + \dfrac{\varphi^{-1}}{2}i = e^{i162} \qquad \lambda_2 = -\dfrac{\sqrt{2+\varphi}}{2} - \dfrac{\varphi^{-1}}{2}i = e^{i198}$

  Such that: $\lambda_1.\lambda_2 = e^{i162}.e^{i198} = 1$

- **For classes P_{3odd} and P_{7even}**

  Characteristic polynomial: $\lambda^2 + \sqrt{2-\varphi^{-1}}\lambda + 1 = 0$

  $\lambda_1 = -\dfrac{\sqrt{2-\varphi^{-1}}}{2} + \dfrac{\varphi}{2}i = e^{i126} \qquad \lambda_2 = -\dfrac{\sqrt{2-\varphi^{-1}}}{2} - \dfrac{\varphi}{2}i = e^{i234}$

  Such that: $\lambda_1.\lambda_2 = e^{i126}.e^{i234} = 1$

- **For classes P_{3even} and P_{7odd}**

  Characteristic polynomial: $\lambda^2 - \sqrt{2-\varphi^{-1}}\lambda + 1 = 0$

  $\lambda_1 = \dfrac{\sqrt{2-\varphi^{-1}}}{2} + \dfrac{\varphi}{2}i = e^{i54} \qquad \lambda_2 = \dfrac{\sqrt{2-\varphi^{-1}}}{2} - \dfrac{\varphi}{2}i = e^{i306}$

  Such that: $\lambda_1.\lambda_2 = e^{i54}.e^{i306} = 1$

Eigenvectors of the golden prime matrix

$\nu_1 = (-i, 1) \qquad \nu_2 = (i, 1)$

Let's look at some examples:

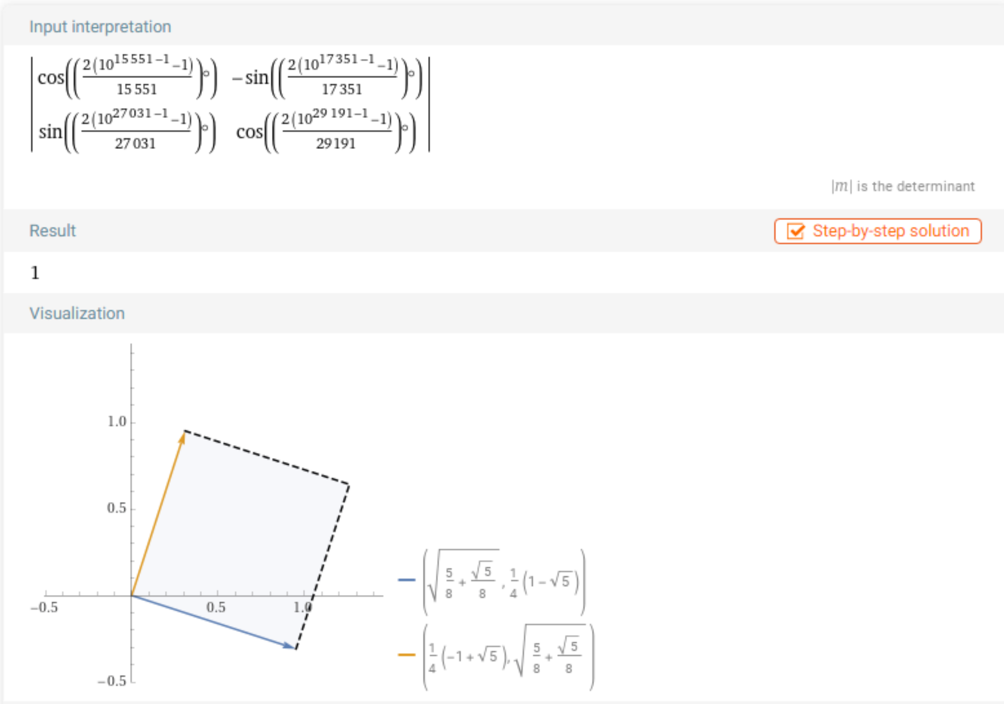

Figure 14: Prime class P_{1odd}.

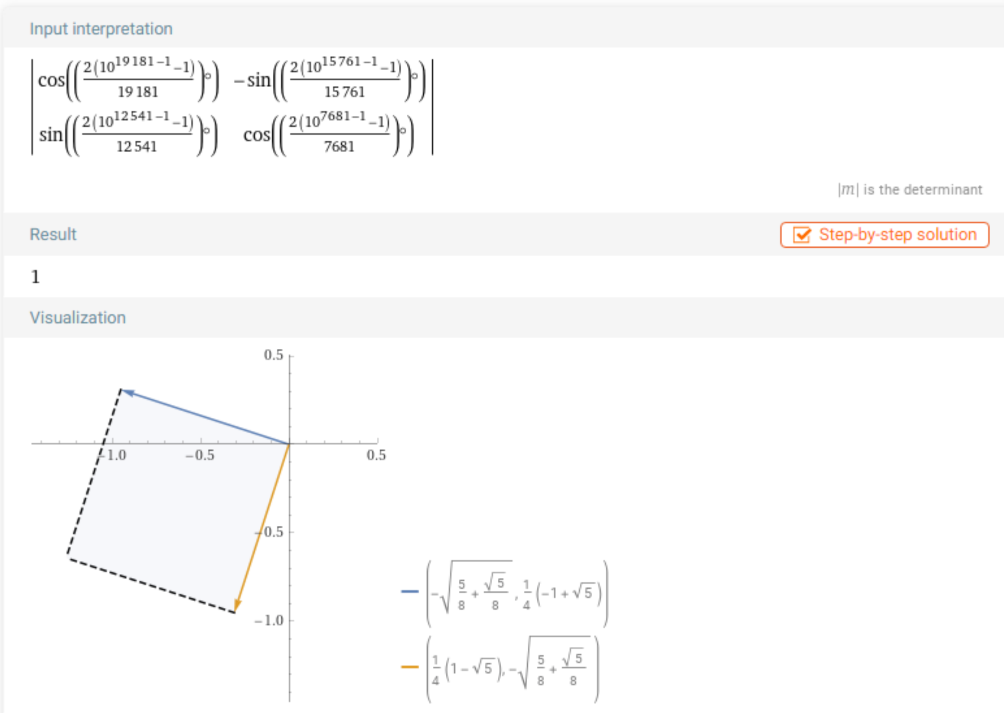

Figure 15: Prime class P_{1even}.

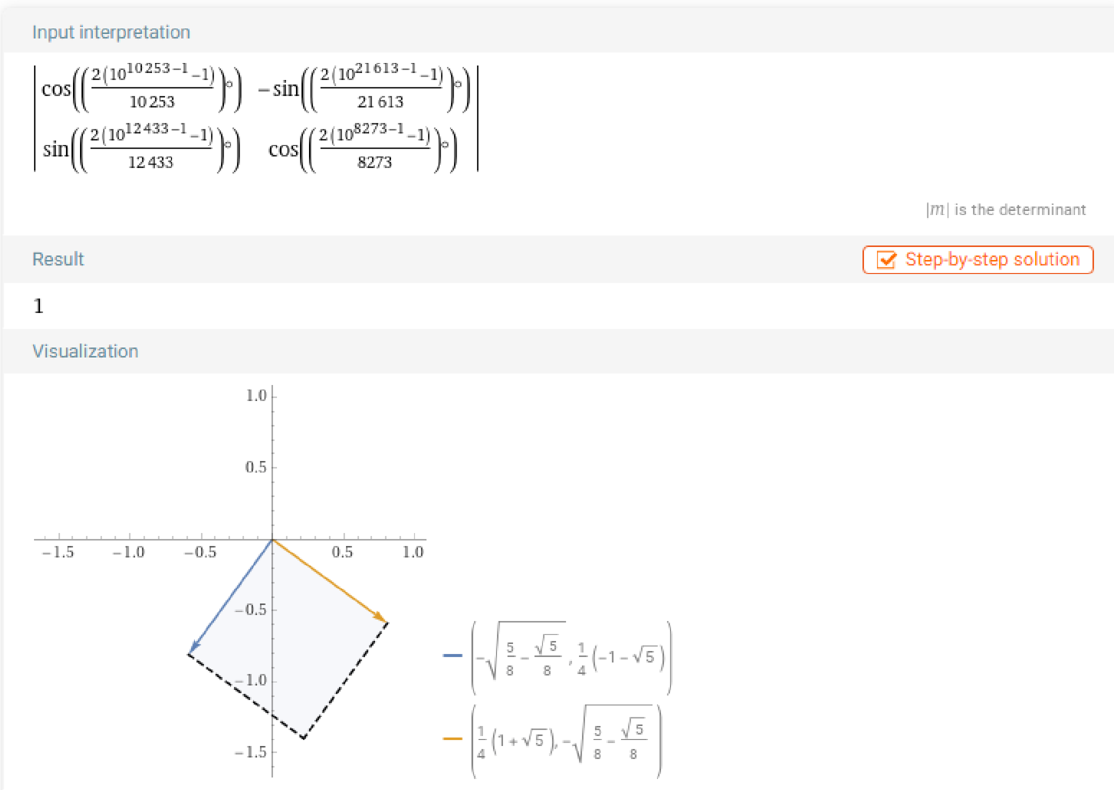

Figure 16: Prime class P_{3odd}.

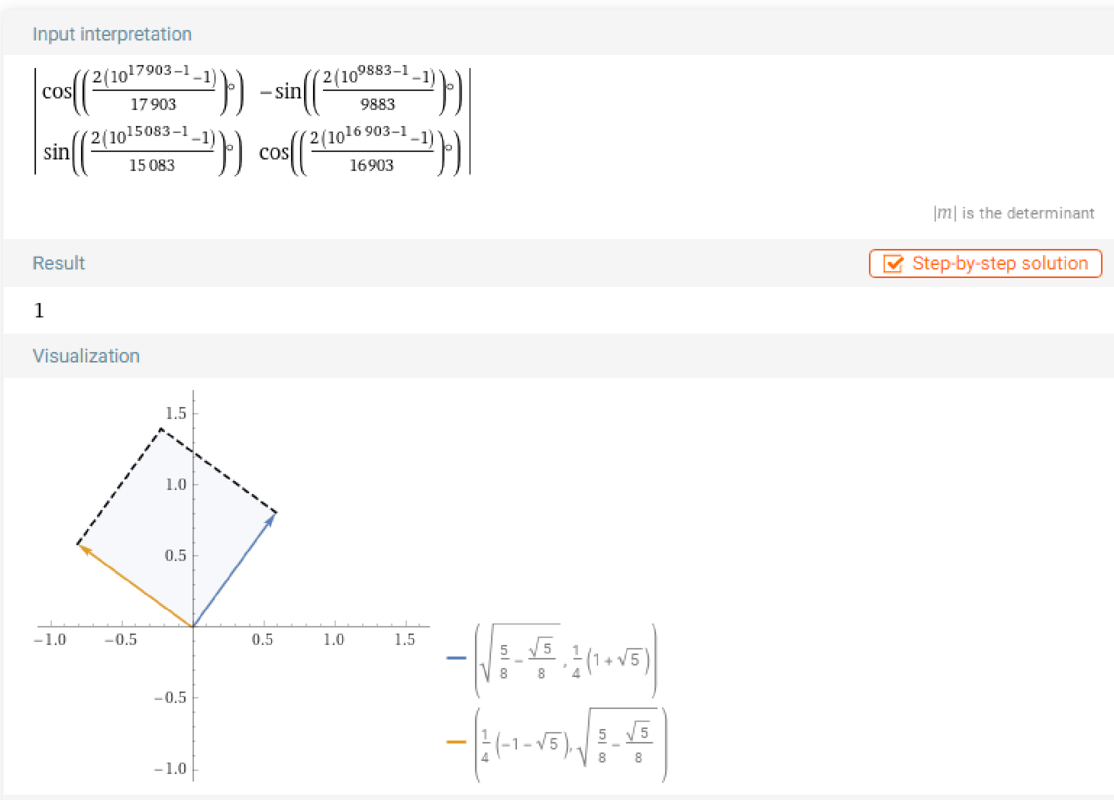

Figure 17: Prime class P_{3even}.

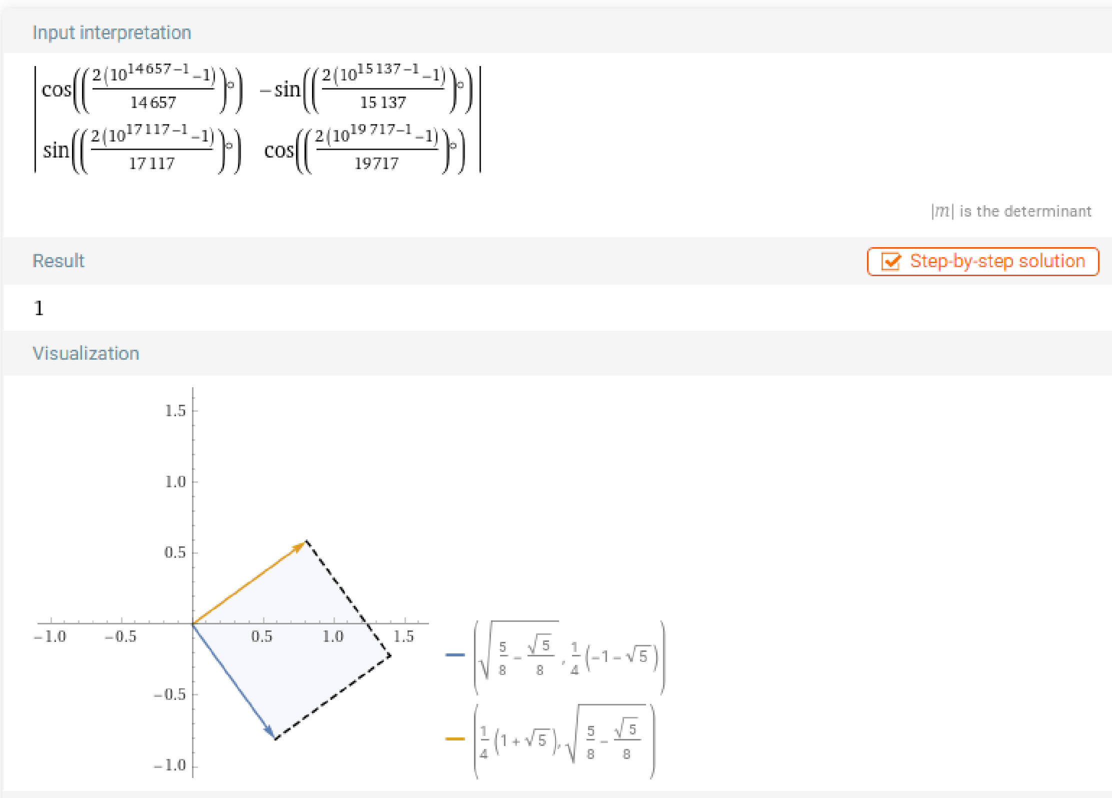

Figure 18: Prime class P_{7odd}.

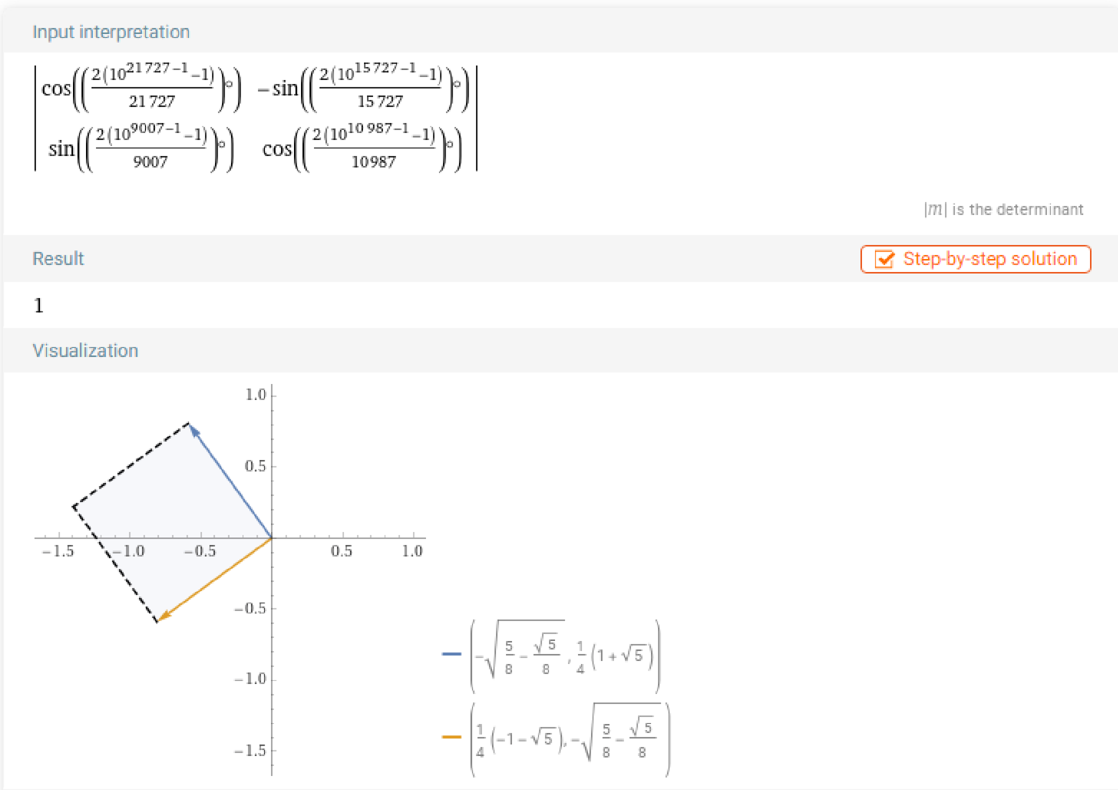

Figure 19: Prime class P_{7even}.

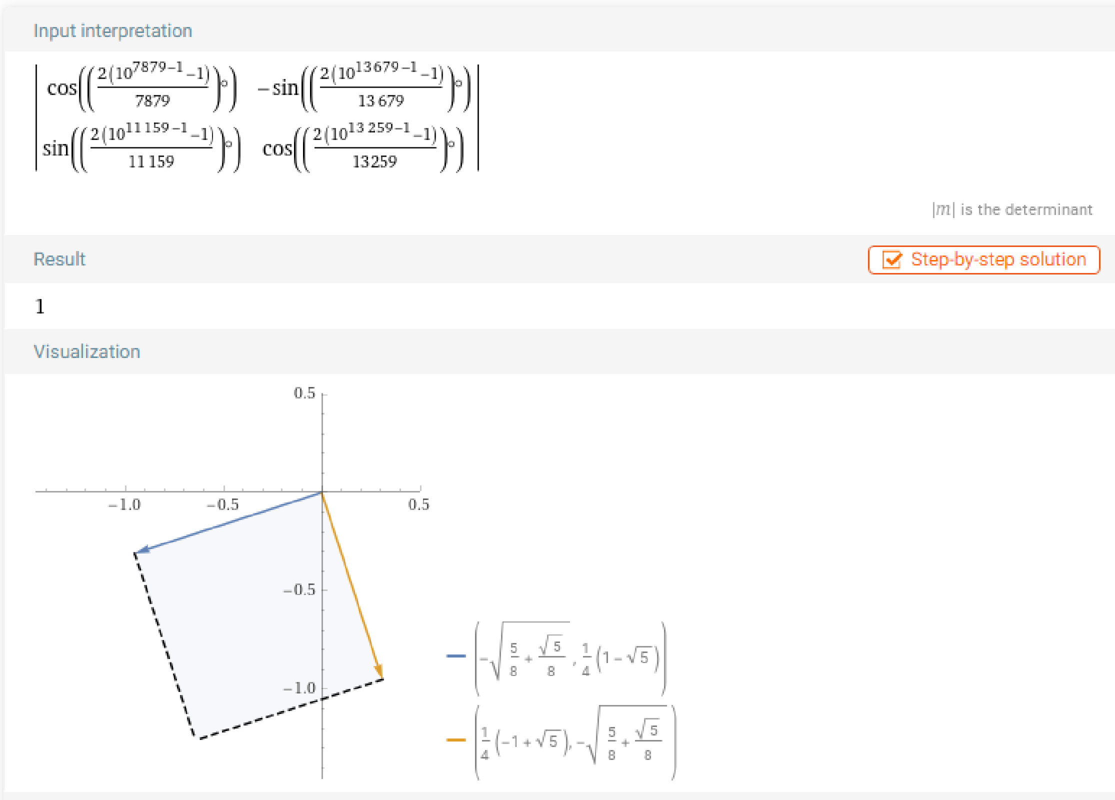

Figure 20: Prime class P_{9odd}.

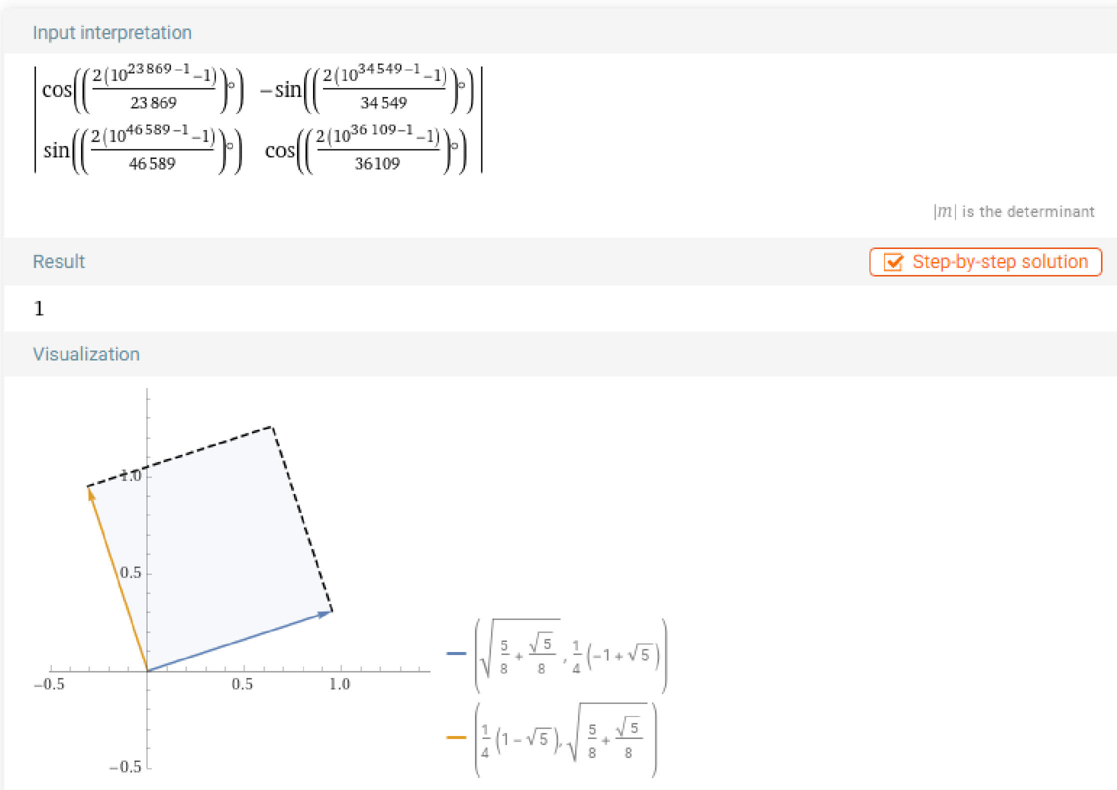

Figure 21: Prime class P_{9even}.

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https://www.wolframalpha.com/input?i=det+%5B+%7B+cos%5B+%282%28%2811047%29%29%5E%

28-1%29%2810%5E%28%2811047%29-1%29+-1%29%29+%5D%C2%B0%2C+-sin%5B+%282%28%2815923%29%29%5E%28-1%29%2810%5E%28%2815923%29-1%29+-1%29%29+%5D+%7D%2C++%7B+sin%5B+%282%28%289887%29%29%5E%28-1%29%2810%5E%28%289887%29-1%29+-1%29%29+%5D+%2C+cos%5B+%282%28%2819867+%29%29%5E%28-1%29%2810%5E%28%2819867+%29-1%29+-1%29%29+%5D%C2%B0+%7D+%5D&lang=es

These golden isometries of prime classes in a three-dimensional matrix rotate as cubes in the space $\mathbb{R}^3$.

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4 Theorem 3: Golden Prime Minimal Polynomial

4.1 Golden Prime Isometries

$\theta = \{18°, 54°, 126°, 162°, 198°, 234°, 306°, 342°\}$

$\theta = \{18°, 54°, 126°, 162°, -162°, -126°, -54°, -18°\}$

$$\theta = \left\{\frac{\pi}{10}, \frac{3\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}, \frac{11\pi}{10}, \frac{13\pi}{10}, \frac{17\pi}{10}, \frac{19\pi}{10}\right\} = \left\{\frac{\pi}{10}, \frac{3\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}, \frac{-9\pi}{10}, \frac{-7\pi}{10}, \frac{-3\pi}{10}, \frac{-\pi}{10}\right\}$$

In the complex plane we have:

$$\begin{aligned}
i^{1/5} &= (-1)^{1/10} = e^{\pi i/10} = e^{i18°} = \cos(18)° + i\sin(18)° \\
i^{3/5} &= (-1)^{3/10} = e^{3\pi i/10} = e^{i54°} = \cos(54)° + i\sin(54)° \\
i^{7/5} &= (-1)^{7/10} = e^{7\pi i/10} = e^{i126°} = \cos(126)° + i\sin(126)° \\
i^{9/5} &= (-1)^{9/10} = e^{9\pi i/10} = e^{i162°} = \cos(162)° + i\sin(162)°
\end{aligned}$$

$$\begin{aligned}
i^{-9/5} &= -i^{1/5} = -(-1)^{1/10} = e^{-9\pi i/10} = e^{i198°} = \cos(198)° + i\sin(198)° \\
i^{-7/5} &= -i^{3/5} = -(-1)^{3/10} = e^{-7\pi i/10} = e^{i234°} = \cos(234)° + i\sin(234)° \\
i^{-3/5} &= -i^{7/5} = -(-1)^{7/10} = e^{-3\pi i/10} = e^{i306°} = \cos(306)° + i\sin(306)° \\
i^{-1/5} &= -i^{9/5} = -(-1)^{9/10} = e^{-\pi i/10} = e^{i342°} = \cos(342)° + i\sin(342)°
\end{aligned}$$ Rotational isometries

taken to the complex plane translate into fifth roots.

$$\begin{aligned}
i^{1/5} &= e^{i18°} = \cos(2p^{-1}(10^{p-1}-1))° + i\sin(2p^{-1}(10^{p-1}-1)° &&\Leftrightarrow p \equiv 11 \pmod{20} \\
i^{3/5} &= e^{i54°} = \cos(2p^{-1}(10^{p-1}-1))° + i\sin(2p^{-1}(10^{p-1}-1)° &&\Leftrightarrow p \equiv 13 \pmod{20} \\
i^{7/5} &= e^{i126°} = \cos(2p^{-1}(10^{p-1}-1)° + i\sin(2p^{-1}(10^{p-1}-1)° &&\Leftrightarrow p \equiv 17 \pmod{20} \\
i^{9/5} &= e^{i162°} = \cos(2p^{-1}(10^{p-1}-1)° + i\sin(2p^{-1}(10^{p-1}-1)° &&\Leftrightarrow p \equiv 19 \pmod{20}
\end{aligned}$$

$$\begin{aligned}
i^{-9/5} &= e^{i198°} = \cos(2p^{-1}(10^{p-1}-1)° + i\sin(2p^{-1}(10^{p-1}-1)° &&\Leftrightarrow p \equiv 1 \pmod{20} \\
i^{-7/5} &= e^{i234°} = \cos(2p^{-1}(10^{p-1}-1)° + i\sin(2p^{-1}(10^{p-1}-1)° &&\Leftrightarrow p \equiv 3 \pmod{20} \\
i^{-3/5} &= e^{i306°} = \cos(2p^{-1}(10^{p-1}-1)° + i\sin(2p^{-1}(10^{p-1}-1)° &&\Leftrightarrow p \equiv 7 \pmod{20} \\
i^{-1/5} &= e^{i342°} = \cos(2p^{-1}(10^{p-1}-1)° + i\sin(2p^{-1}(10^{p-1}-1)° &&\Leftrightarrow p \equiv 9 \pmod{20}
\end{aligned}$$

For a radius $r = 1$ and the angles $\theta = \{18°, 54°, 126°, 162°, 198°, 234°, 306°, 342°\}$

where $\varphi = \dfrac{1+\sqrt{5}}{2}$

$$z_1 = e^{i18} = \cos(18)^\circ + i\sin(18)^\circ = \frac{\sqrt{2+\varphi}}{2} + \frac{\varphi^{-1}}{2}i$$
$$z_2 = e^{i54} = \cos(54)^\circ + i\sin(54)^\circ = \frac{\sqrt{2-\varphi^{-1}}}{2} + \frac{\varphi}{2}i$$
$$z_3 = e^{i126} = \cos(126)^\circ + i\sin(126)^\circ = -\frac{\sqrt{2-\varphi^{-1}}}{2} + \frac{\varphi}{2}i$$
$$z_4 = e^{i162} = \cos(162)^\circ + i\sin(162)^\circ = -\frac{\sqrt{2+\varphi}}{2} + \frac{\varphi^{-1}}{2}i$$
$$z_5 = e^{i198} = \cos(198)^\circ + i\sin(198)^\circ = -\frac{\sqrt{2+\varphi}}{2} - \frac{\varphi^{-1}}{2}i$$
$$z_6 = e^{i234} = \cos(234)^\circ + i\sin(234)^\circ = -\frac{\sqrt{2-\varphi^{-1}}}{2} - \frac{\varphi}{2}i$$
$$z_7 = e^{i306} = \cos(306)^\circ + i\sin(306)^\circ = \frac{\sqrt{2-\varphi^{-1}}}{2} - \frac{\varphi}{2}i$$
$$z_8 = e^{i342} = \cos(342)^\circ + i\sin(342)^\circ = \frac{\sqrt{2+\varphi}}{2} - \frac{\varphi^{-1}}{2}i$$

So the product of the conjugates is 1

$$\Big(\cos(18)^\circ + i\sin(18)^\circ\Big)\Big(\cos(342)^\circ + i\sin(342)^\circ\Big) = 1$$
$$\Big(\cos(54)^\circ + i\sin(54)^\circ\Big)\Big(\cos(306)^\circ + i\sin(306)^\circ\Big) = 1$$
$$\Big(\cos(126)^\circ + i\sin(126)^\circ\Big)\Big(\cos(234)^\circ + i\sin(234)^\circ\Big) = 1$$
$$\Big(\cos(162)^\circ + i\sin(162)^\circ\Big)\Big(\cos(198)^\circ + i\sin(198)^\circ\Big) = 1$$

Let's look at some examples:

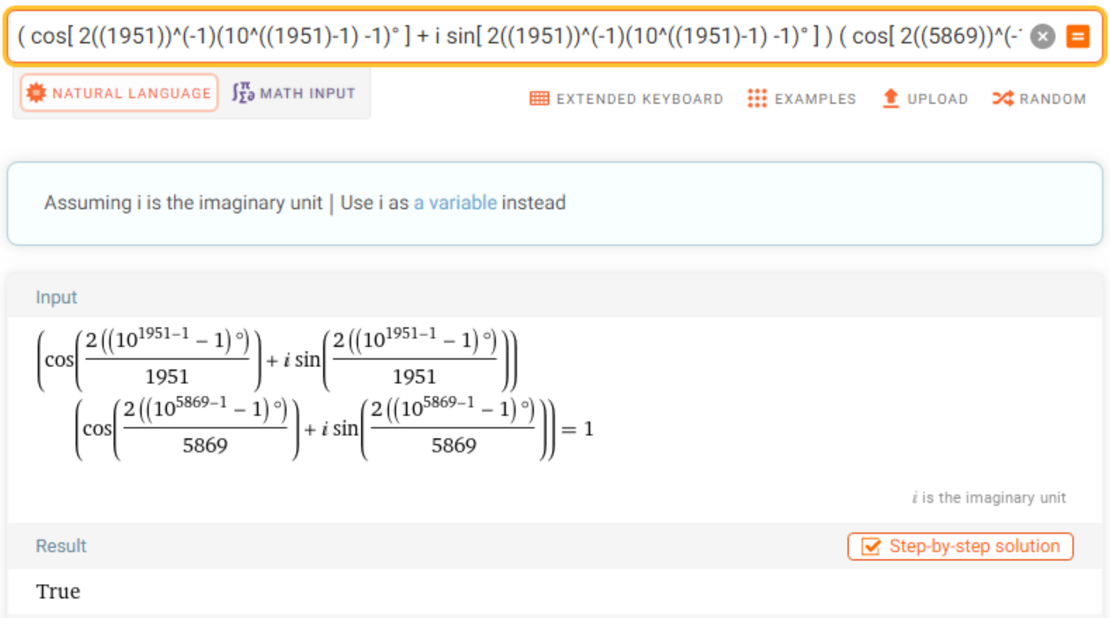

Figure 22: Conjugate prime classes P_{1odd} and P_{9even}.

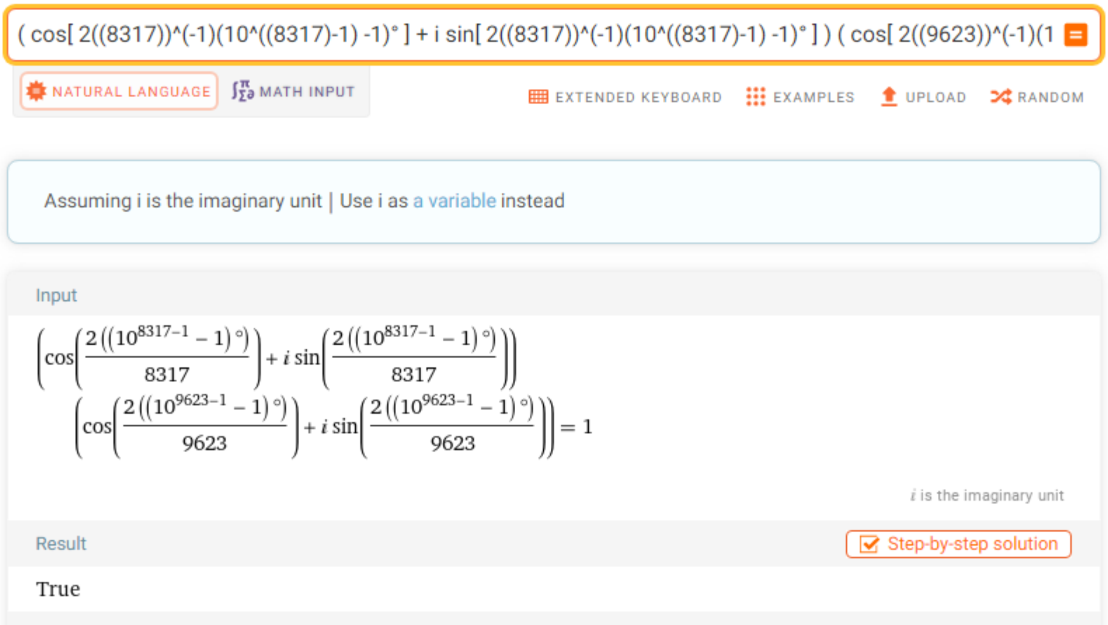

Figure 23: Conjugate prime classes P_{3even} and P_{7odd}.

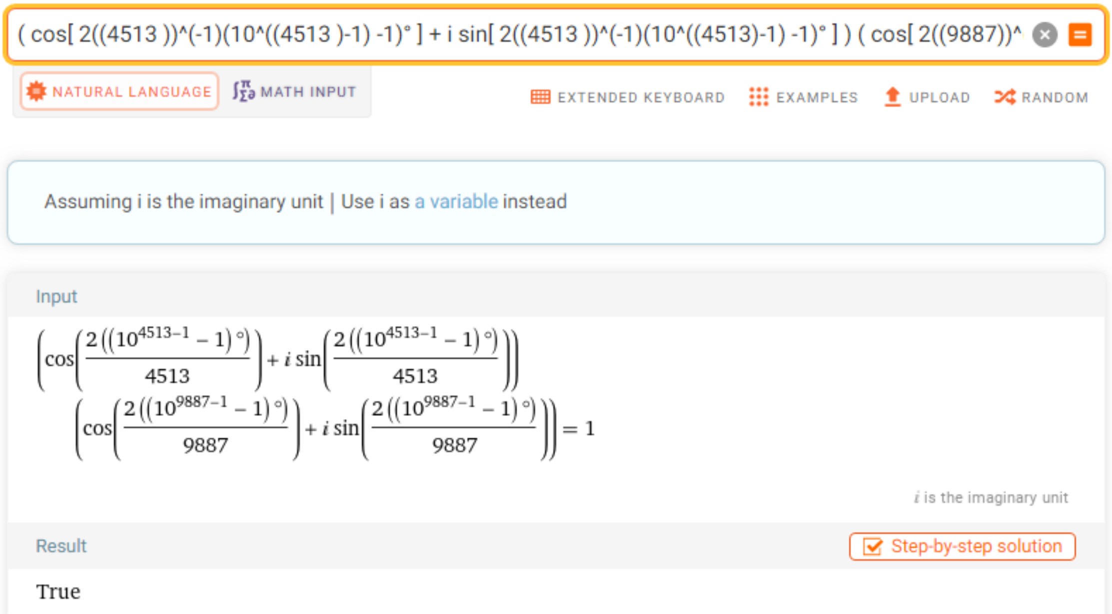

Figure 24: Conjugate prime classes P_{3odd} and P_{7even}.

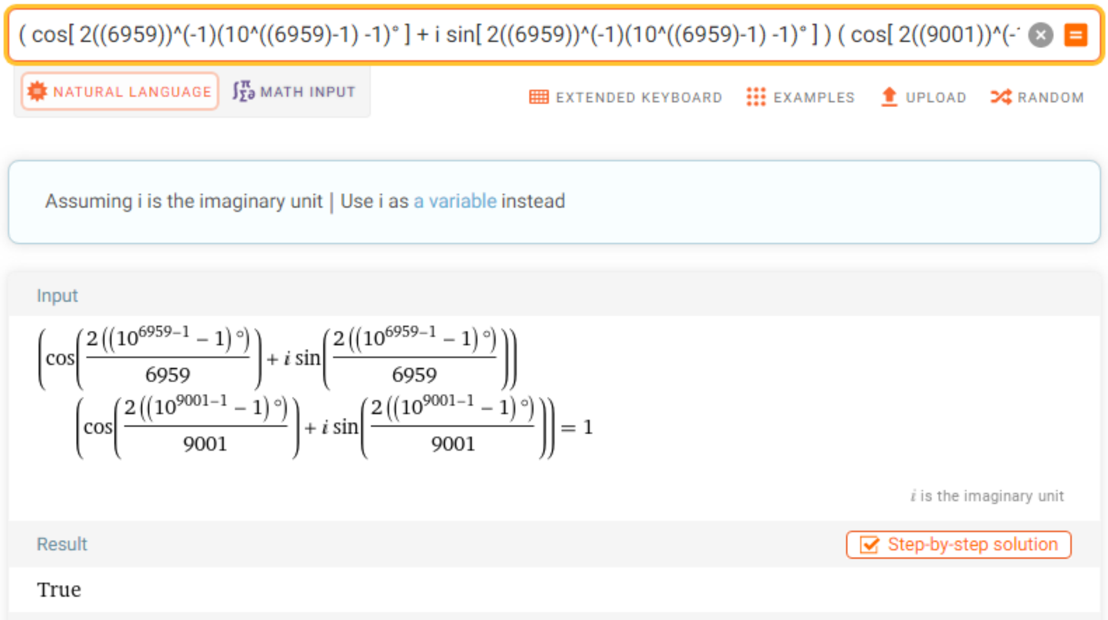

Figure 25: Conjugate prime classes P_{1even} and P_{9odd}.

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4.2 Golden Prime Cyclotomic Polynomial

The 8 golden prime isometries in the complex plane correspond to the 8 roots of unity of the following minimal polynomial:

$$\sum_{n=0}^{4}(-1)^n x^{2n} = x^8 - x^6 + x^4 - x^2 + 1 = 0$$

$$x^8 - x^6 + x^4 - x^2 + 1 =$$
$$= (x - i^{1/5})(x - i^{3/5})(x - i^{7/5})(x - i^{9/5})(x - i^{-1/5})(x - i^{-3/5})(x - i^{-7/5})(x - i^{-9/5})$$
$$= (x - e^{\pi i/10})(x - e^{3\pi i/10})(x - e^{7\pi i/10})(x - e^{9\pi i/10})(x - e^{-\pi i/10})(x - e^{-3\pi i/10})(x - e^{-7\pi i/10})(x - e^{-9\pi i/10})$$

https://www.wolframalpha.com/input?i=x%5E8+-+x%5E6+%2B+x%5E4+-+x%5E2+%2B1+%3D+%28x-i%5E%281%2F5%29+%29%28x-i%5E%283%2F5%29+%29%28x-i%5E%287%2F5%29+%29%28x-i%5E%289%2F5%29+%29%28x%E2%88%92i%5E%28-1%2F5%29+%29%28x%E2%88%92i%5E%28-3%2F5%29%29%28x%E2%88%92i%5E%28-7%2F5%29+%29%28x%E2%88%92i%5E%28-9%2F5%29%29

4.3 Golden Prime Integral

The integral of the minimal polynomial whose zeros are the 8 prime classes has an interesting result:

$$\int_0^\infty = \frac{dx}{x^8 - x^6 + x^4 - x^2 + 1} = \frac{\pi}{\sqrt{5}} = \frac{\pi}{\varphi + \varphi^{-1}}$$

Thus, a new identity of π in terms of the 8 infinite prime classes is:

$$\pi = (\varphi + \varphi^{-1}) \int_0^\infty \frac{dx}{x^8 - x^6 + x^4 - x^2 + 1}$$

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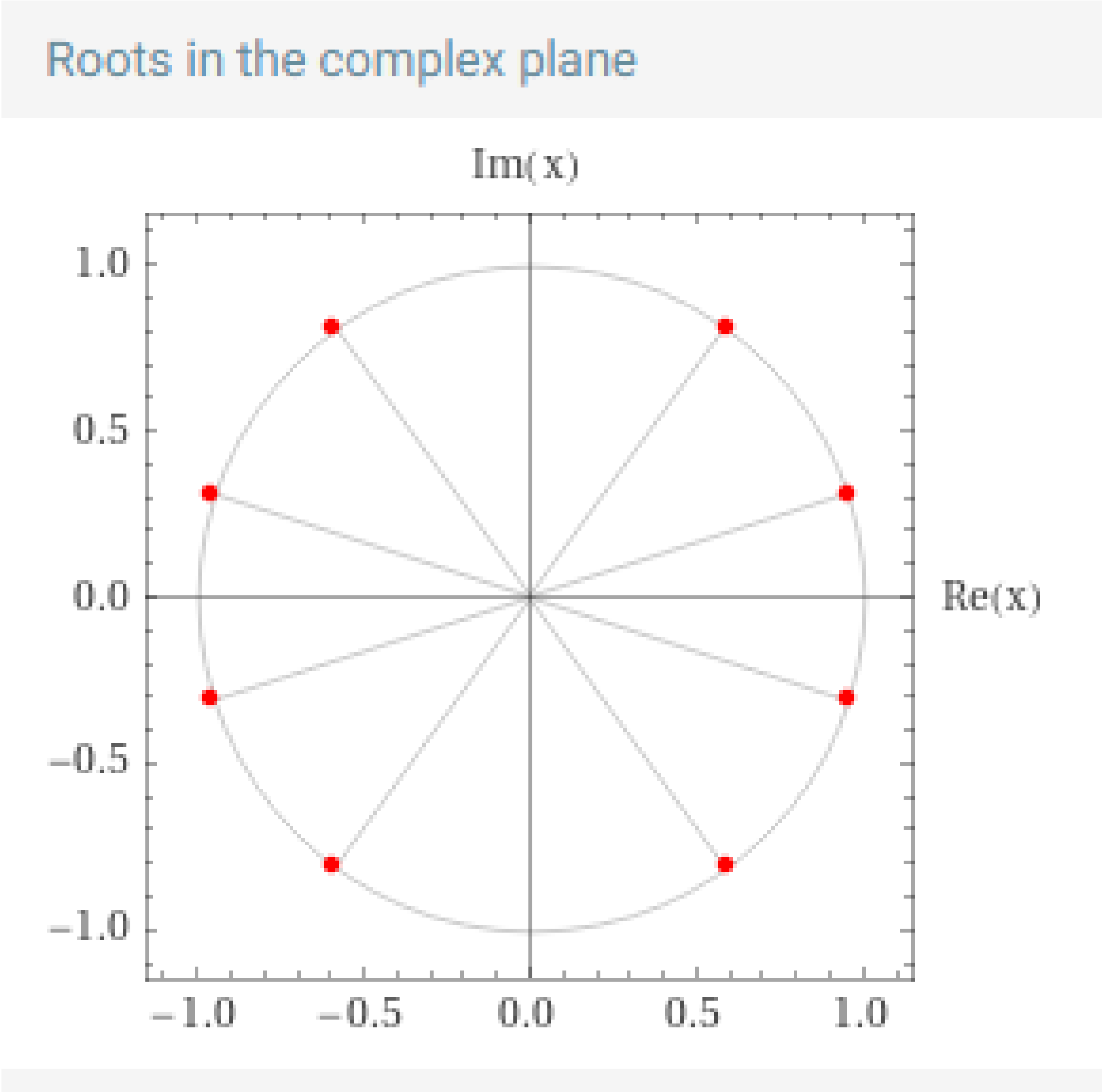

Figure 26: Golden prime classes in the complex plane.

5 Theorem of Golden Prime Exponential

The exponential function can be expressed as the sum of the hyperbolic sine and the hyperbolic cosine, $e^x = \sinh(x) + \cosh(x)$. Taking $x = 2\sin(2p^{-1}(10^{p-1} - 1))^\circ$ and let $p > 5$, $q > 5$ be prime, we obtain 8 invariant images and ordered by the prime residual classes.

$$f(p,q) = \sinh\left(2\sin(2p^{-1}(10^{p-1} - 1))^\circ\right) + \cosh\left(2\sin(2q^{-1}(10^{q-1} - 1))^\circ\right) =$$

$$a = e^{\varphi}, \qquad b = e^{-\varphi}, \qquad c = e^{1/\varphi}, \qquad d = e^{-1/\varphi}$$

$$k = \tfrac{1}{2}(-e^{\varphi} + e^{-\varphi} + e^{1/\varphi} + e^{-1/\varphi}), \qquad f = \tfrac{1}{2}(e^{\varphi} - e^{-\varphi} + e^{1/\varphi} + e^{-1/\varphi})$$

$$g = \tfrac{1}{2}(e^{\varphi} + e^{-\varphi} - e^{1/\varphi} + e^{-1/\varphi}), \qquad h = \tfrac{1}{2}(e^{\varphi} + e^{-\varphi} + e^{1/\varphi} - e^{-1/\varphi})$$

	P_{1odd}	P_{3odd}	P_{7odd}	P_{9odd}	P_{1even}	P_{3even}	P_{7even}	P_{9even}
P_{1odd}	c	h	h	c	c	h	h	c
P_{3odd}	f	a	a	f	f	a	a	f
P_{7odd}	f	a	a	f	f	a	a	f
P_{9odd}	c	h	h	c	c	h	h	c
P_{1even}	d	g	g	d	d	g	g	d
P_{3even}	k	b	b	k	k	b	b	k
P_{7even}	k	b	b	k	k	b	b	k
P_{9even}	d	g	g	d	d	g	g	d

Figure 27: Table of symmetrical combinations of prime classes.

Proof

If $x = \varphi$, then: $e^{\varphi} = \sinh(\varphi) + \cosh(\varphi)$. The same for the rest of the cases.

By theorem 1 one has that the primes partition into 8 equivalence classes modulo 360 as isometries. Therefore, under the exponential function they become golden powers of the number e. Permutation between the prime classes yields a linear combination of exponential golden powers. $\square$

Example:

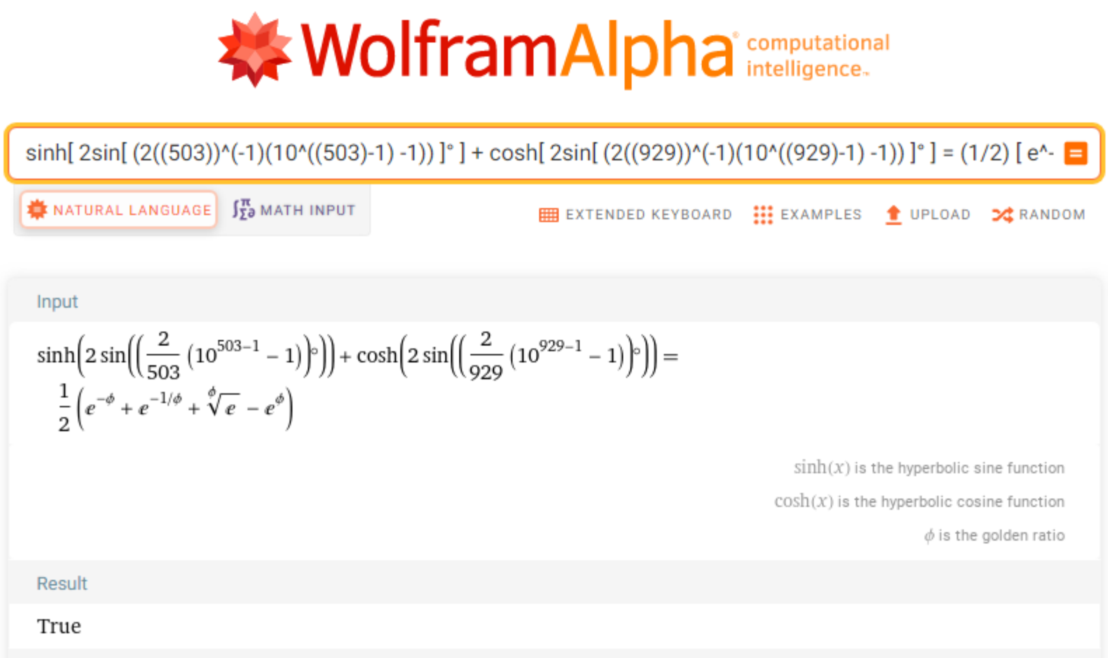

Figure 28: Example for classes P_{3even} y P_{9even} in WolframAlpha.

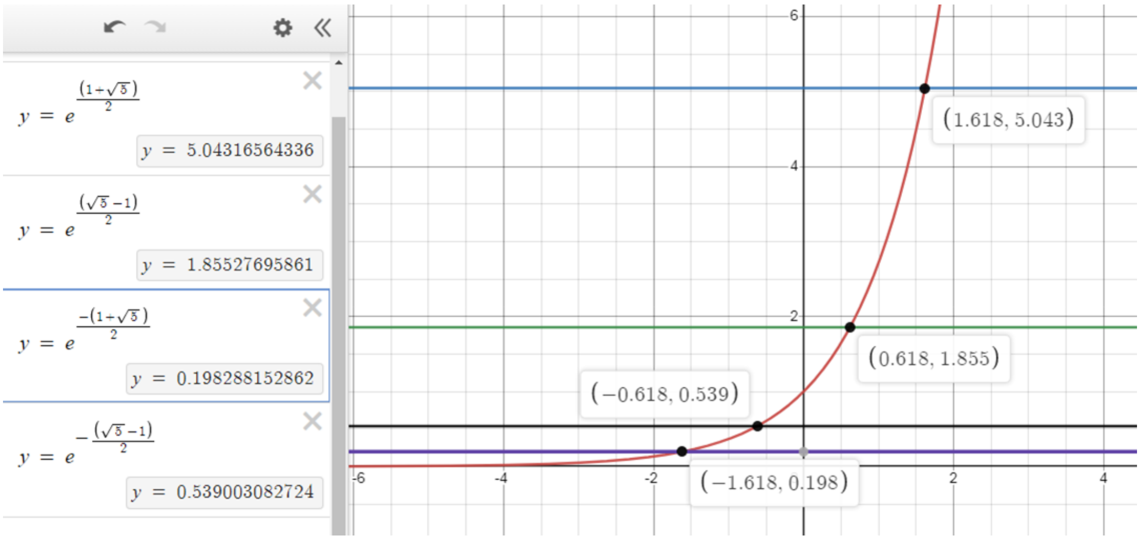

Figure 29: Golden exponential points in Geogebra.

https://www.wolframalpha.com/input?i=sinh%5B+2sin%5B+%282%28%284621%29%29%5E%28-1%29%2810%5E%28%284621%29-1%29+-1%29%29+%5D%C2%B0+%5D+%2B+cosh%5B+2sin%5B+%282%28%285233%29%29%5E%28-1%29%2810%5E%28%285233%29-1%29+-1%29%29+%5D%C2%B0+%5D+%3D+%281%2F2%29+%5B+e%5E-%CF%86+%2B+e%5E%28-1%2F%CF%86%29+-+e%5E%281%2F%CF%86%29+%2B+e%5E%CF%86+%5D&lang=es

6 Theorem of Golden Prime Hiperbolic Quotient

The quotient of the hyperbolic sine by the hyperbolic cosine of the prime golden function yields 8 golden exponential images which are the various combinations of the prime classes. Let $p > 5$, $q > 5$ prime is satisfied:

$$f(p,q) = \frac{\sinh(\sin(2p^{-1}(10^{p-1}-1))^{\circ}}{\cosh(\sin(2q^{-1}(10^{q-1}-1))^{\circ}} =$$

$$a = \frac{e^{\varphi}-e^{-\varphi}}{e^{\varphi}+e^{-\varphi}}, \qquad b = \frac{e^{-\varphi}-e^{\varphi}}{e^{\varphi}+e^{-\varphi}}, \qquad c = \frac{e^{-\varphi}-e^{\varphi}}{e^{1/\varphi}+e^{-1/\varphi}}, \qquad d = \frac{e^{\varphi}-e^{-\varphi}}{e^{1/\varphi}+e^{-1/\varphi}}$$

$$e = \frac{e^{1/\varphi}-e^{-1/\varphi}}{e^{\varphi}+e^{-\varphi}}, \qquad f = \frac{e^{-1/\varphi}-e^{1/\varphi}}{e^{\varphi}+e^{-\varphi}}, \qquad g = \frac{e^{1/\varphi}-e^{-1/\varphi}}{e^{1/\varphi}+e^{-1/\varphi}}, \qquad h = \frac{e^{-1/\varphi}-e^{1/\varphi}}{e^{1/\varphi}+e^{-1/\varphi}}$$

For the prime classes the symmetrical combinations are:

	P_{1odd}	P_{3odd}	P_{7odd}	P_{9odd}	P_{1even}	P_{3even}	P_{7even}	P_{9even}
P_{1odd}	g	e	e	g	g	e	e	g
P_{3odd}	d	a	a	d	d	a	a	d
P_{7odd}	d	a	a	d	d	a	a	d
P_{9odd}	g	e	e	g	g	e	e	g
P_{1even}	h	f	f	h	h	f	f	h
P_{3even}	c	b	b	c	c	b	b	c
P_{7even}	c	b	b	c	c	b	b	c
P_{9even}	h	f	f	h	h	f	f	h

Figure 30: Table of symmetrical combinations of prime classes.

Proof

Since the prime classes are invariant angles it follows that:

$\sinh(\varphi) = \frac{1}{2}(e^{\varphi} - e^{-\varphi}) \qquad \cosh(\varphi) = \frac{1}{2}(e^{\varphi} + e^{-\varphi})$

Then the quotient is: $\dfrac{e^{\varphi} - e^{-\varphi}}{e^{\varphi} + e^{-\varphi}}$

And for the rest of the cases the same. Thus, the combinations in the table are given. $\square$

Example:

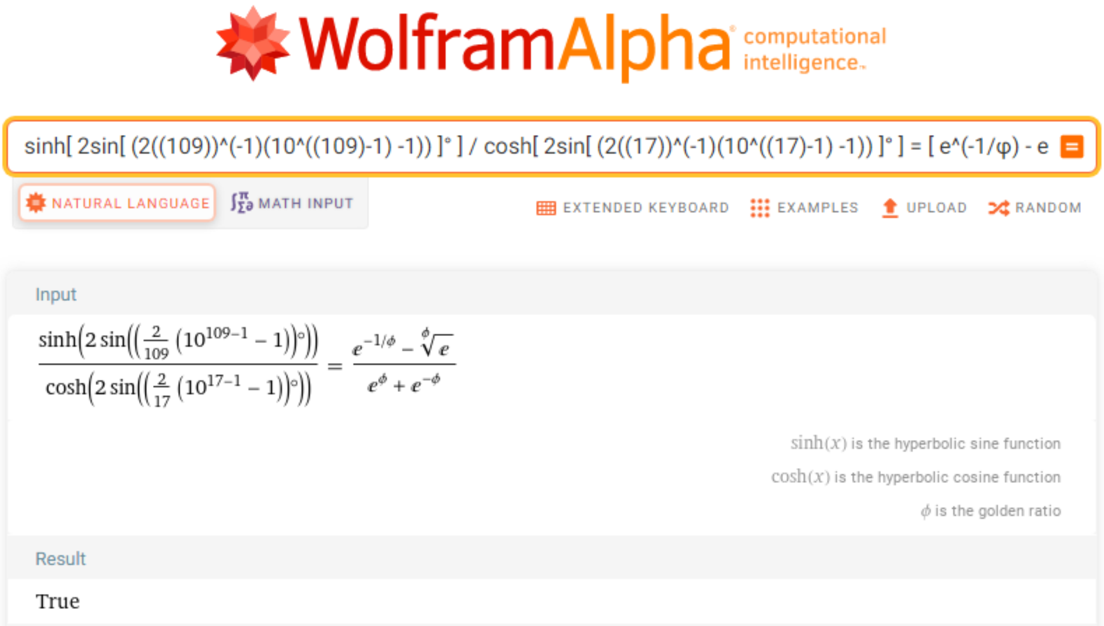

Figure 31: Example for classes P_{7odd} and P_{9even}.

https://www.wolframalpha.com/input?i=sinh%5B+2sin%5B+%282%28%28911%29%29%5E%28-1%29%2810%5E%28%28911%29-1%29+-1%29%29+%5D%C2%B0+%5D+%2F+cosh%5B+2sin%5B+%282%28%28311%29%29%5E%28-1%29%2810%5E%28%28311%29-1%29+-1%29%29+%5D%C2%B0+%5D+%3D++%5B+e%5E%281%2F%CF%86%29+-+e%5E%28-1%2F%CF%86%29+%5D+%2F+%5B+e%5E%281%2F%CF%86%29+%2B+e%5E%28-1%2F%CF%86%29+%5D&lang=es

7 Digital Structure of Fibonacci Sequence

The majestic Fibonacci sequence from the point of view of the digital roots of each element is nothing more than a cyclic loop of 24 digits that iterate over and over in a definite and well-ordered pattern. In fact, what is most interesting is that dividing the 24-digit loop into two blocks of 12 elements and adding them term by term all add up to 9 with its mirror-image complementary. The digital root of an integer is the sum of its digits until a single digit is obtained. So if we obtain the digital root of each element of the Fibonacci sequence we get the following:

The cycle that repeats every 24 terms of the sequence is:

$1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9$

1	1	2	3	5	8	4	3	7	1	8	9
8	8	7	6	4	1	5	6	2	8	1	9
9	9	9	9	9	9	9	9	9	9	9	9

Figure 32: Table of Fibonacci's sum of digital roots.

n	F_n	Digital root	n	F_n	Digital root
1	1	1	13	233	8
2	1	1	14	377	8
3	2	2	15	610	7
4	3	3	16	987	6
5	5	5	17	1597	4
6	8	8	18	2584	1
7	13	4	19	4181	5
8	21	3	20	6765	6
9	34	7	21	10946	2
10	55	1	22	17711	8
11	89	8	23	28657	1
12	144	9	24	46368	9

n	F_n	Digital root	n	F_n	Digital root
25	75025	1	37	24157817	8
26	121393	1	38	39088169	8
27	196418	2	39	63245986	7
28	317811	3	40	102334155	6
29	514229	5	41	165580141	4
30	832040	8	42	267914296	1
31	1346269	4	43	433494437	5
32	2178309	3	44	701408733	6
33	3524578	7	45	1134903170	2
34	5702887	1	46	1836311903	8
35	9227465	8	47	2971215073	1
36	14930352	9	48	48075269768	9

Figure 33: Table of Fibonacci digital roots.

$F_1 \equiv 1 \pmod 9$	$F_{13} \equiv 8 \pmod 9$
$F_2 \equiv 1 \pmod 9$	$F_{14} \equiv 8 \pmod 9$
$F_3 \equiv 2 \pmod 9$	$F_{15} \equiv 7 \pmod 9$
$F_4 \equiv 3 \pmod 9$	$F_{16} \equiv 6 \pmod 9$
$F_5 \equiv 5 \pmod 9$	$F_{17} \equiv 4 \pmod 9$
$F_6 \equiv 8 \pmod 9$	$F_{18} \equiv 1 \pmod 9$
$F_7 \equiv 4 \pmod 9$	$F_{19} \equiv 5 \pmod 9$
$F_8 \equiv 3 \pmod 9$	$F_{20} \equiv 6 \pmod 9$
$F_9 \equiv 7 \pmod 9$	$F_{21} \equiv 2 \pmod 9$
$F_{10} \equiv 1 \pmod 9$	$F_{22} \equiv 8 \pmod 9$
$F_{11} \equiv 8 \pmod 9$	$F_{23} \equiv 1 \pmod 9$
$F_{12} \equiv 0 \pmod 9$	$F_{24} \equiv 9 \pmod 9$

Figure 34: Table of classes modulo 9 of Fibonacci.

7.1 Theorem of Twelfth Fibonacci

The twelfth Fibonacci numbers are divisible by 9.

Proof

$F_{12n} \equiv 0 \pmod 9 \Leftrightarrow F_{12n-1} \equiv F_{12n+1} \pmod 9$

We know that $F_{12n-1} + F_{12n} = F_{12n+1}$, luego $F_{12n-1} + F_{12n} \equiv F_{12n+1} \pmod 9$

by hypothesis, $F_{12n} \equiv 0 \pmod 9$, luego $F_{12n-1} \equiv F_{12n+1} \pmod 9$

reciprocally, $F_{12n} = F_{12n+1} - F_{12n-1}$, then $F_n \equiv F_{12n+1} - F_{12n-1} \pmod 9 \equiv 0 \pmod 9$

since by hypothesis $F_{12n-1} \equiv F_{12n+1} \pmod 9$. Therefore, $9|F_{12k}$ □

8 Theorem 5: Golden Fibonacci Partition

The golden ratio is the cosine image of the discrete space of a subsuccession of the Fibonacci sequence, namely, the twelfth Fibonacci numbers, i.e., those found at every 12th term of the sequence. And the modulus 360° *yields a partition into* 8 *infinite families or Fibonacci classes that are invariant under the isometries of the regular pentagon and its opposite rotation.*

g: $\mathbb{F}_{12k} \to \mathbb{R}$

$k \longmapsto g(k)$

$$g(k) = 2\cos(F_{12k})^\circ = \begin{cases} \varphi & si \quad F_{12k(4)} - 180, \qquad F_{12k(6)} - 180 \\ \varphi^{-1} & si \quad F_{12k(2)}, \qquad F_{12k(8)} \\ -\varphi & si \quad F_{12k(4)}, \qquad F_{12k(6)} \\ -\varphi^{-1} & si \quad F_{12k(2)} - 180, \qquad F_{12k(8)} - 180 \end{cases}$$

Where: $\varphi = \dfrac{1+\sqrt{5}}{2}$; $\quad \varphi^{-1} = \dfrac{\sqrt{5}-1}{2}$

There exists a partition into equivalence classes of the F_{12k} with $k \in \mathbb{Z}$, corresponding to the isometries of the regular pentagon:

$F_{12k(4)} \equiv 144 \pmod{360} \quad \Leftrightarrow k \equiv (1,6) \pmod{10}$ whose last digit ends in 4

$F_{12k(8)} \equiv 288 \pmod{360} \quad \Leftrightarrow k \equiv (2,7) \pmod{10}$ whose last digit ends in 8

$$
\begin{aligned}
&F_{12k(2)} \equiv 72 \pmod{360} && \Leftrightarrow k \equiv (3,8) \pmod{10} && \text{whose last digit ends in } 2\\
&F_{12k(6)} \equiv 216 \pmod{360} && \Leftrightarrow k \equiv (4,9) \pmod{10} && \text{whose last digit ends in } 6
\end{aligned}
$$

$$
\begin{aligned}
&F_{12k(4)} - 180 \equiv 324 \pmod{360} && \Leftrightarrow k \equiv (1,6) \pmod{10} && \text{whose last digit ends in } 4\\
&F_{12k(8)} - 180 \equiv 108 \pmod{360} && \Leftrightarrow k \equiv (2,7) \pmod{10} && \text{whose last digit ends in } 8\\
&F_{12k(2)} - 180 \equiv 252 \pmod{360} && \Leftrightarrow k \equiv (3,8) \pmod{10} && \text{whose last digit ends in } 2\\
&F_{12k(6)} - 180 \equiv 36 \pmod{360} && \Leftrightarrow k \equiv (4,9) \pmod{10} && \text{whose last digit ends in } 6
\end{aligned}
$$

In general, it is complied with:

$$
\begin{aligned}
&2\cos(144)^\circ = 2\cos(F_{12k(4)})^\circ = -\varphi && \Leftrightarrow k \equiv (1,6) \pmod{10}\\
&2\cos(288)^\circ = 2\cos(F_{12k(8)})^\circ = \varphi^{-1} && \Leftrightarrow k \equiv (2,7) \pmod{10}\\
&2\cos(72)^\circ \ = 2\cos(F_{12k(2)})^\circ = \varphi^{-1} && \Leftrightarrow k \equiv (3,8) \pmod{10}\\
&2\cos(216)^\circ = 2\cos(F_{12k(6)})^\circ = -\varphi && \Leftrightarrow k \equiv (4,9) \pmod{10}
\end{aligned}
$$

And for the rotated pentagon 180° we have:

$$
\begin{aligned}
&2\cos(324)^\circ = 2\cos(F_{12k(4)} - 180)^\circ = \varphi && \Leftrightarrow k \equiv (1,6) \pmod{10}\\
&2\cos(108)^\circ = 2\cos(F_{12k(8)} - 180)^\circ = -\varphi^{-1} && \Leftrightarrow k \equiv (2,7) \pmod{10}\\
&2\cos(252)^\circ = 2\cos(F_{12k(2)} - 180)^\circ = -\varphi^{-1} && \Leftrightarrow k \equiv (3,8) \pmod{10}\\
&2\cos(36)^\circ \ = 2\cos(F_{12k(6)} - 180)^\circ = \varphi && \Leftrightarrow k \equiv (4,9) \pmod{10}
\end{aligned}
$$

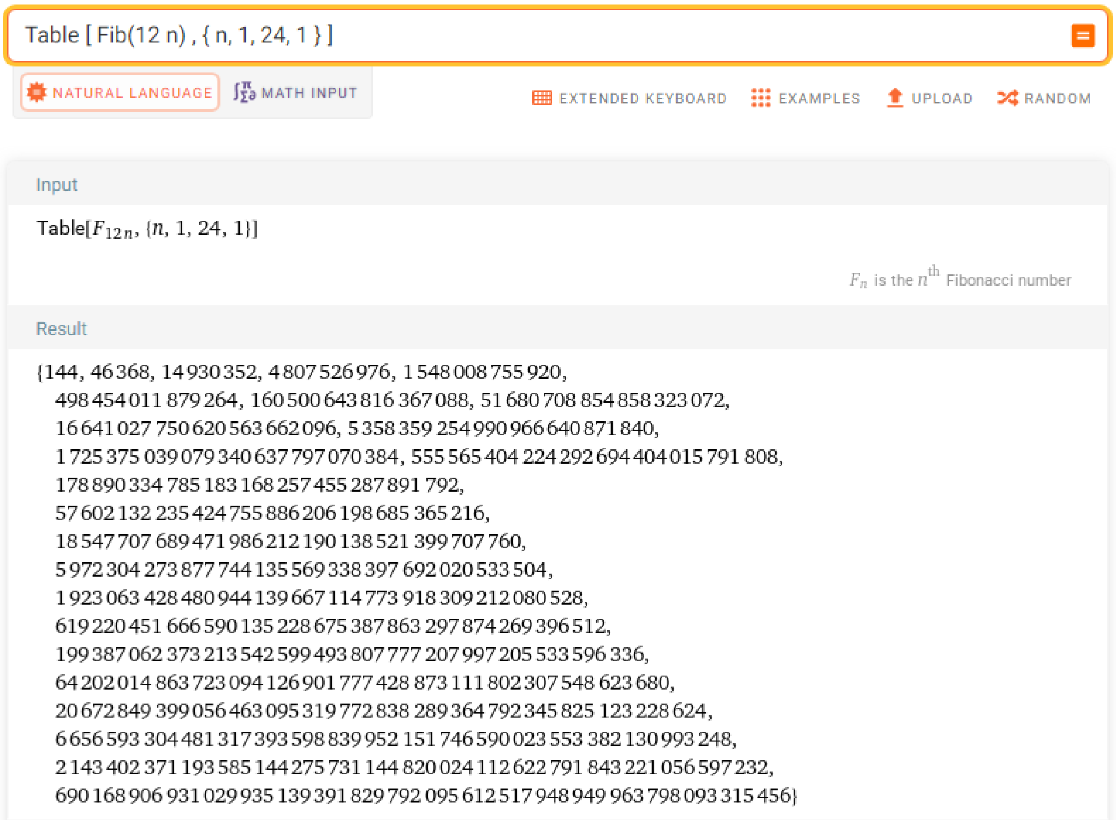

Figure 35: Table of the first 24 twelfths Fibonacci numbers.

In this table we see the first 24 twelfth Fibonacci numbers. And in the next two tables are the module 360 classes of the F_{12k} and $F_{12k} - 180$, which correspond to the isometries of the regular pentagon and its opposite rotation. There are 8 residual classes of the Fibonacci twelfths. The 0 class and the 180 class are automatically excluded, because their cosine image is not the golden ratio. The 8 residual classes of both opposite pentagons are: $\{72°, 144°, 216°, 288°\}$ and $\{36°, 108°, 252°, 324°\}$.

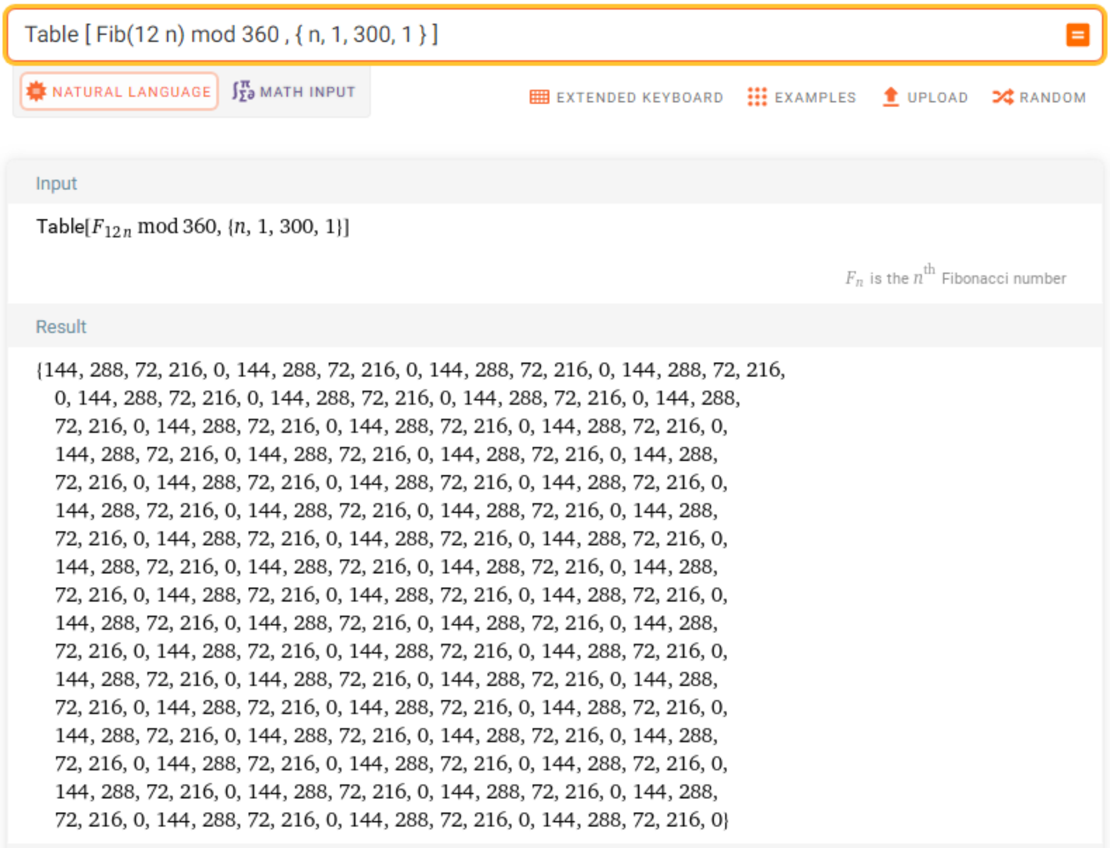

Figure 36: Table of classes modulo 360 of twelfths Fibonacci numbers.

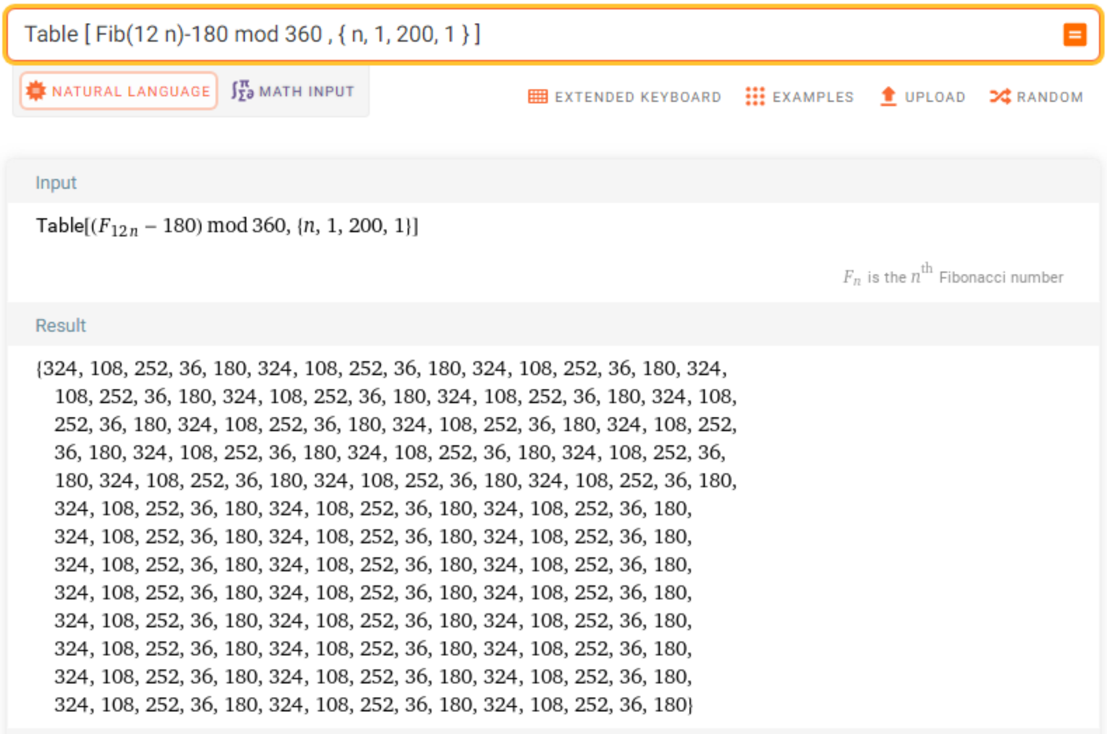

Figure 37: Table of classes modulo 360 of twelfths Fibonacci numbers rotated 180.

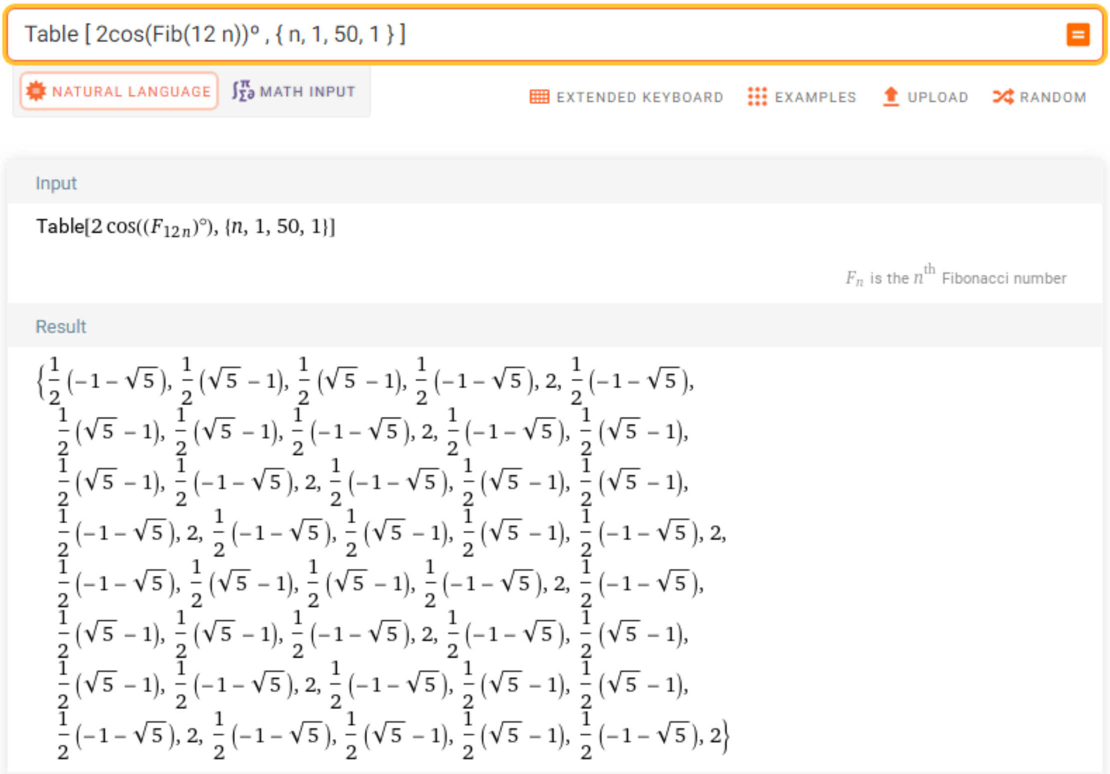

Figure 38: Table of the first 50 twelfth Fibonacci and its cosine golden image.

https://www.wolframalpha.com/input?i=Table+%5B+2cos%28Fib%2812+n%29%29%C2%B0+%2C+ +%7B+n%2C+1%2C+200%2C+1+%7D+%5D&lang=es

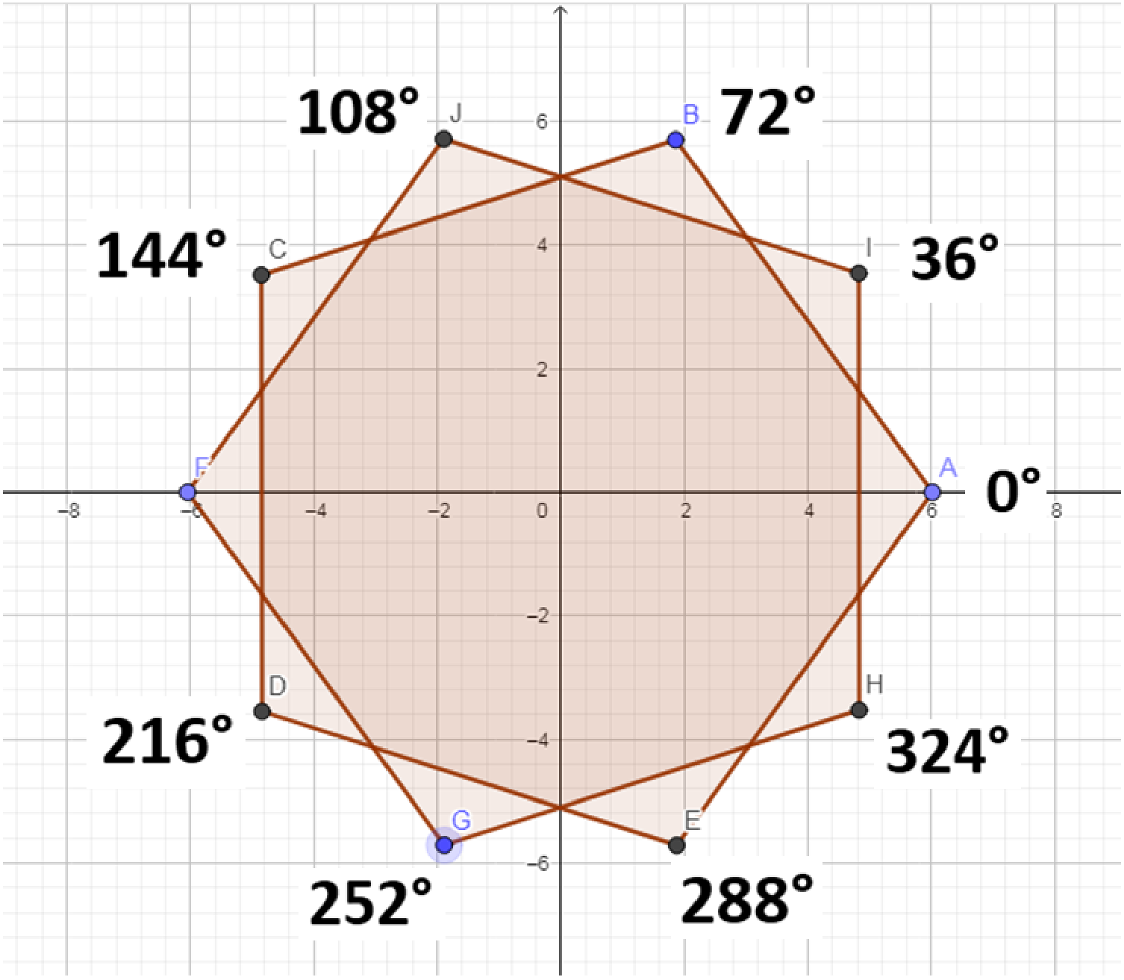

Figure 39: Twelfths classes of Fibonacci.

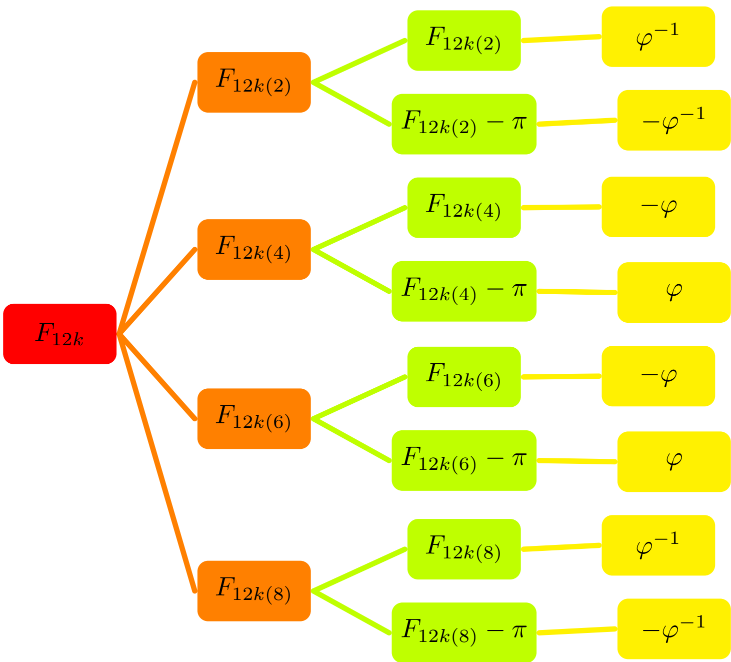

Figure 40: Golden Fibonacci tree.

8.1 Golden Fibonacci Equivalence Relation

There exists an equivalence relation $\sim$ on the set of Fibonacci twelfths such that it determines a partition into 8 residual classes modulo 360.

The 8 Fibonacci golden classes are:

$F_{12k(2)} \equiv 72 \pmod{360}$ $\quad F_{12k(4)} \equiv 144 \pmod{360}$

$F_{12k(6)} \equiv 216 \pmod{360}$ $\quad F_{12k(8)} \equiv 288 \pmod{360}$

$F_{12k(2)} - 180 \equiv 252 \pmod{360}$ $\quad F_{12(4)} - 180 \equiv 324 \pmod{360}$

$F_{12k(6)} - 180 \equiv 36 \pmod{360}$ $\quad F_{12k(8)} - 180 \equiv 108 \pmod{360}$

The 8 residual Fibonacci classes are golden angles. Since the angles are invariant for each class, it is satisfied that the 8 classes are disjoint and their union is equal to the entire infinite set of twelfth Fibonacci numbers F_{12k}, except the 0 class and the 180 class, because their cosine image is not the golden ratio.

$F_{12k(i)} \cap F_{12k(j)} = \emptyset$ $\qquad \bigcup F_{12k(i)} = F_{12k}$ where $i = 2, 4, 6, 8$

Quotient set

$F_{12k}/\sim = \{[72°], [144°], [216°], [288°]\}$

$(F_{12k} - 180)/\sim = \{[36°], [108°], [252°], [324°]\}$

Proposition

The partition of the Fibonacci twelfths for $k \in \mathbb{Z}$ occurs in the finite field $\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$. Recall that the class of 0 is excluded, because its cosine image is not the golden ratio.

$\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$.

So that: $F_{12k} = F_{12\mathbb{Z}_5}$

$k = \{5n + 0,\ 5n + 1,\ 5n + 2,\ 5n + 3,\ 5n + 4\}$

$F_{12k(0)} = F_{12(5n+0)}$ that are $F_{12k} \equiv 0 \pmod{10}$

$F_{12k(4)} = F_{12(5n+1)}$ that are: $F_{12k} \equiv 4 \pmod{10}$

$F_{12k(8)} = F_{12(5n+2)}$ that are: $F_{12k} \equiv 8 \pmod{10}$

$F_{12k(2)} = F_{12(5n+3)}$ that are: $F_{12k} \equiv 2 \pmod{10}$

$F_{12k(6)} = F_{12(5n+4)}$ that are: $F_{12k} \equiv 6 \pmod{10}$

Now let's look at the 3 properties of every equivalence relation:

Reflexive: $\forall x \in F_{12k} : x \sim x$

i.e. Sea $F_{12} \in F_{12(5n+1)}$ y $F_{12} \sim F_{12}$

Simetric: $\forall x, y \in F_{12k} : x \sim y \Rightarrow y \sim x$

i.e. Sean $F_{12} \in F_{12(5n+1)}$, $\quad F_{72} \in F_{12(5n+1)}$: $\quad F_{12} \sim F_{72} \Rightarrow F_{72} \sim F_{12}$

Transitive: $\forall x, y, z \in F_{12k} : x \sim y \wedge y \sim z \Rightarrow x \sim z$

i.e. Let be $F_{12} \in F_{12(5n+1)}$, $\quad F_{72} \in F_{12(5n+1)}$, $\quad F_{132} \in F_{12(5n+1)}$:

$F_{12} \sim F_{72} \wedge F_{72} \sim F_{132} \Rightarrow F_{12} \sim F_{132}$ $\square$

Input

Table[$F_{12(5n+1)}$, {n, 0, 10, 1}]

F_n is the n^{th} Fibonacci number

Result

n	$F_{12(5n+1)}$
0	144
1	498 454 011 879 264
2	1 725 375 039 079 340 637 797 070 384
3	5 972 304 273 877 744 135 569 338 397 692 020 533 504
4	20 672 849 399 056 463 095 319 772 838 289 364 792 345 825 123 228 624
5	71 558 092 601 766 452 430 641 106 302 905 217 344 934 236 440 122 960 ⋱ 529 002 115 744
6	247 694 960 571 651 628 711 444 594 884 429 646 292 615 632 415 916 575 ⋱ 771 902 992 555 242 690 154 864
7	857 384 416 798 688 205 322 146 546 398 461 722 061 337 599 142 249 183 ⋱ 459 012 869 301 255 270 820 177 904 036 305 984
8	2 967 795 697 064 976 427 862 421 836 334 141 336 150 900 792 786 444 ⋱ 618 288 545 455 102 252 664 927 332 007 624 673 682 829 385 529 104
9	10 272 884 749 181 815 606 072 318 598 530 637 999 272 870 421 445 820 ⋱ 271 435 224 647 698 308 362 799 217 540 057 689 492 273 456 451 819 361 ⋱ 258 784 224

Figure 41: Table of class $F_{12k(4)} = F_{12(5n+1)}$.

https://www.wolframalpha.com/input?i=Table+%5B+Fib%2812+%285n%2B1%29+%29+%2C++%7B+n%2C+0%2C+10%2C+1+%7D+%5D

8.2 Twelfths Fibonacci classes.

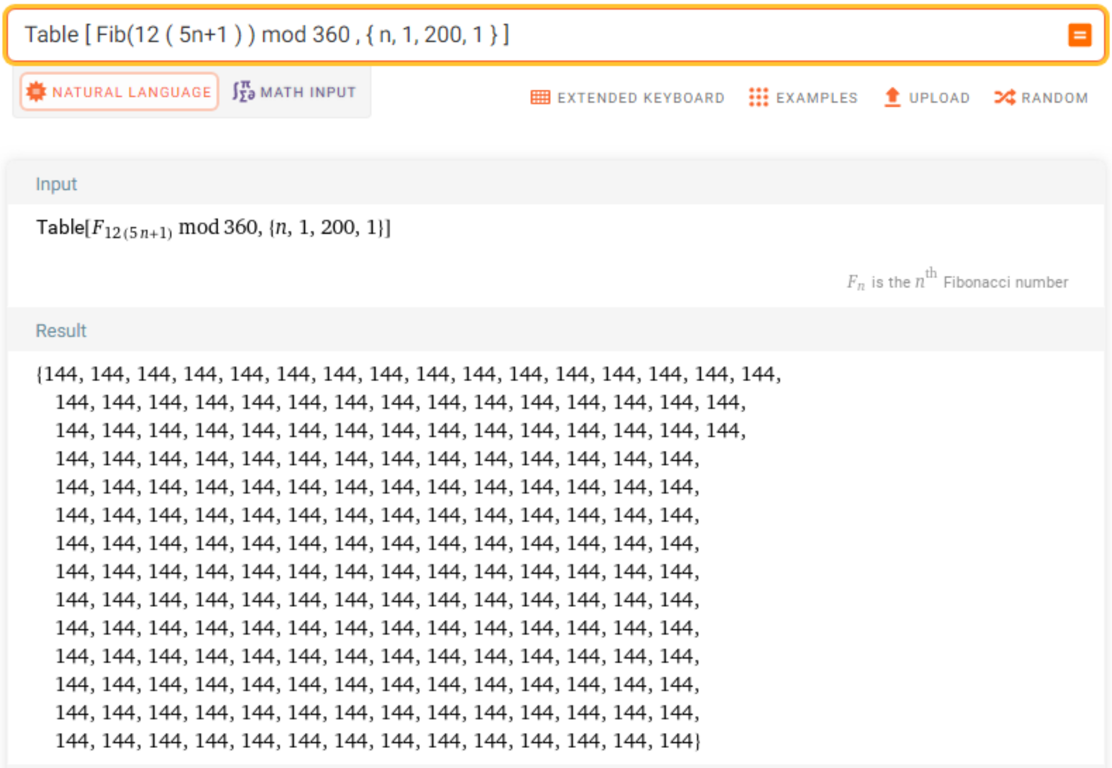

Figure 42: Table of class $F_{12k(4)}$.

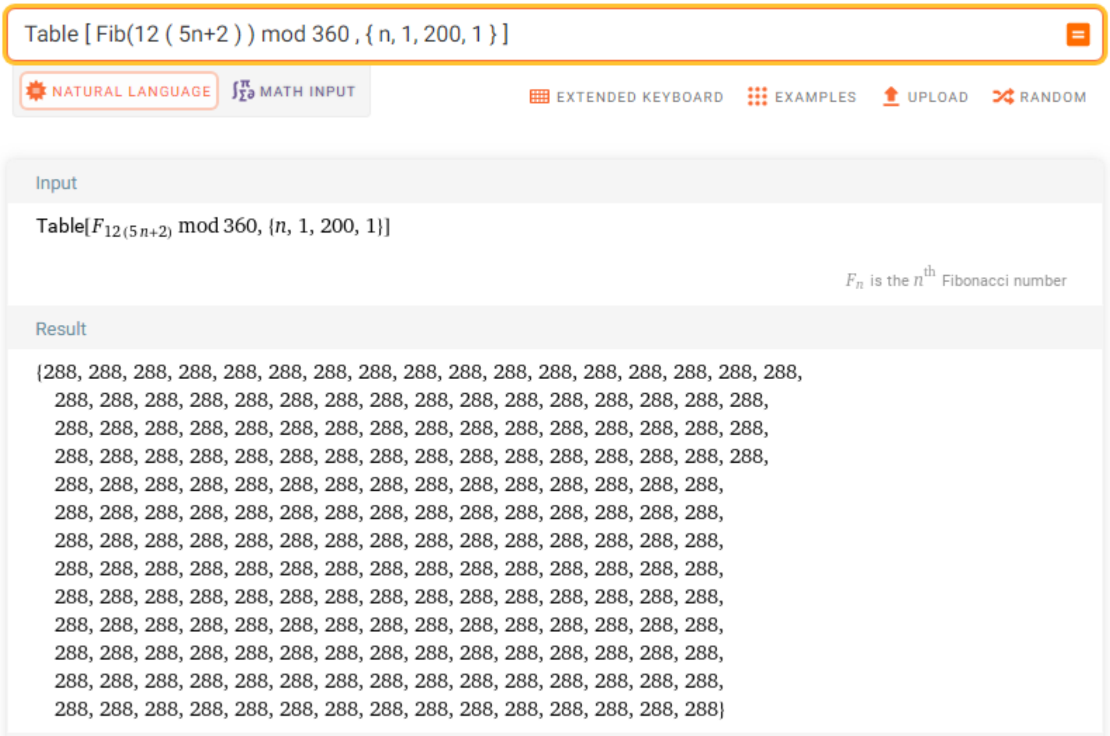

Figure 43: Table of class $F_{12k(8)}$.

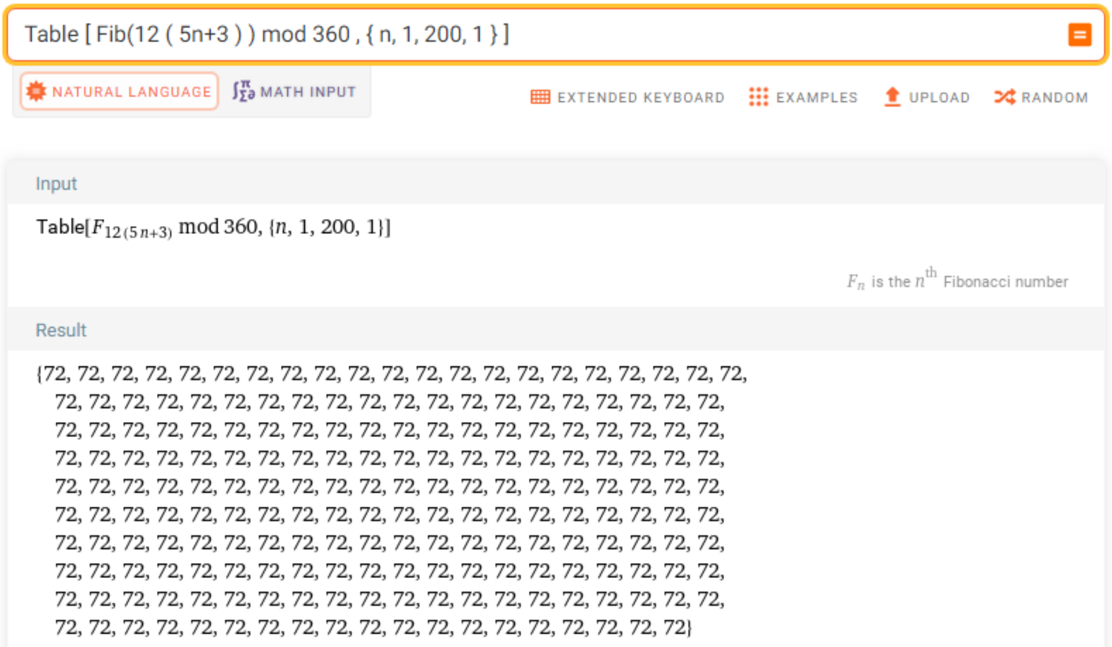

Figure 44: Table of class $F_{12k(2)}$.

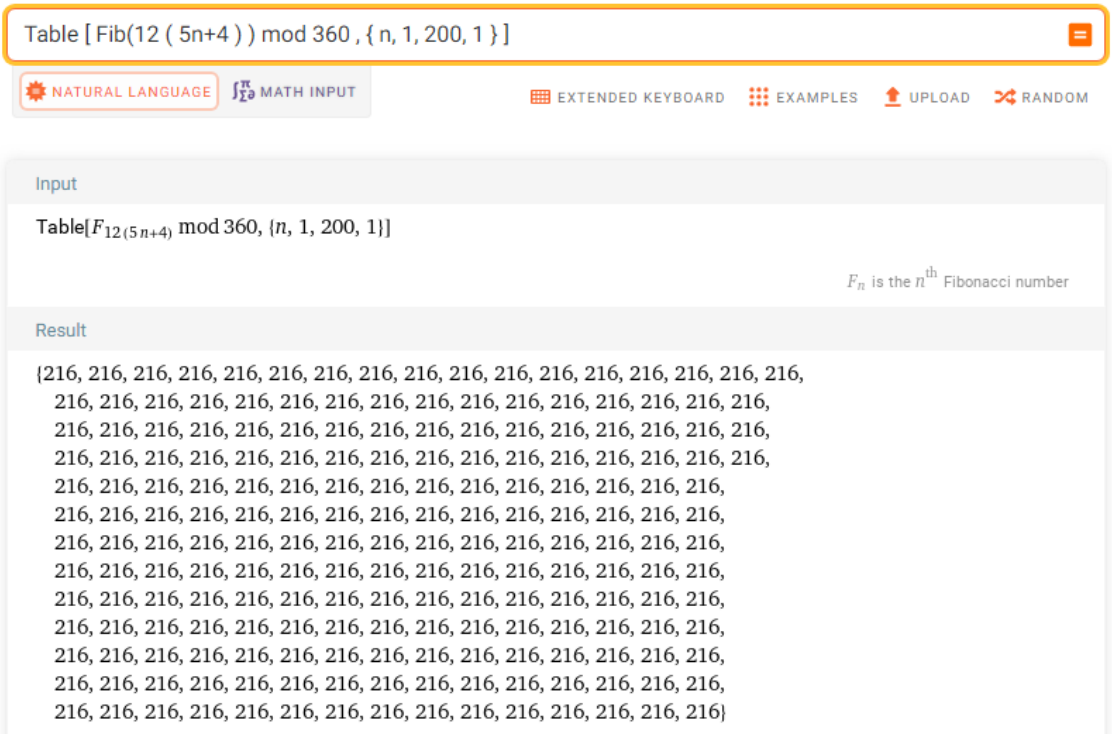

Figure 45: Table of class $F_{12k(6)}$.

8.3 Twelfths Fibonacci rotated classes.

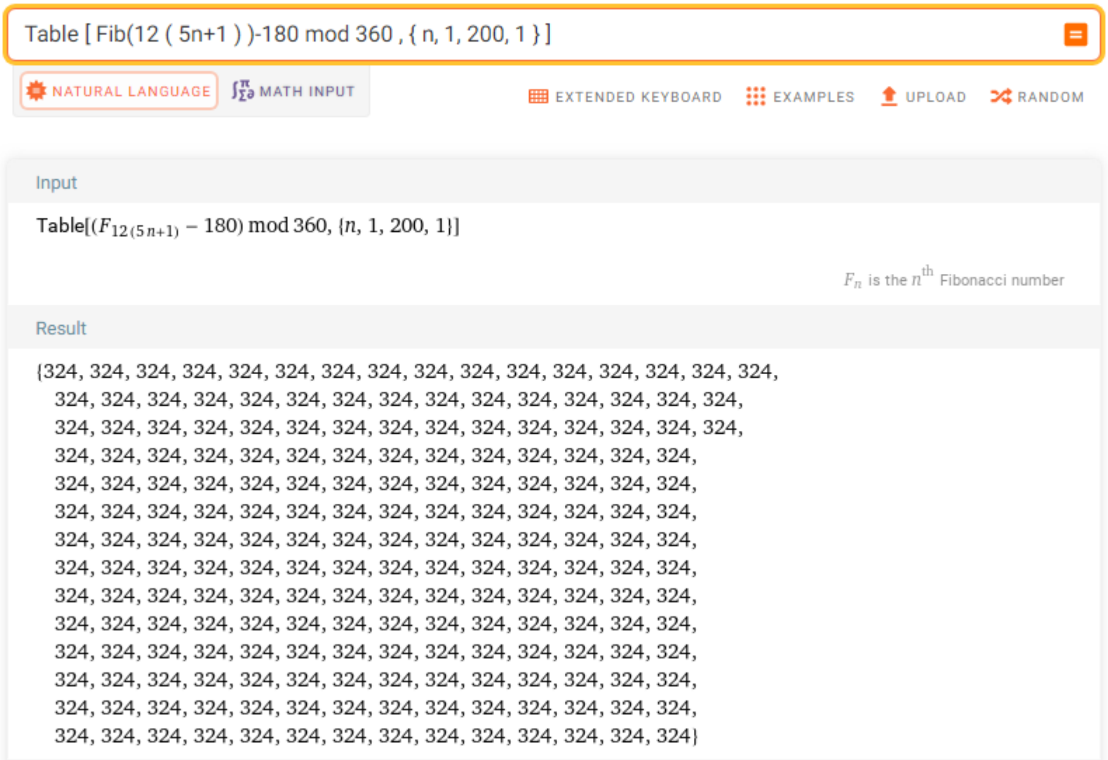

Figure 46: Table of class $F_{12k(4)} - 180$.

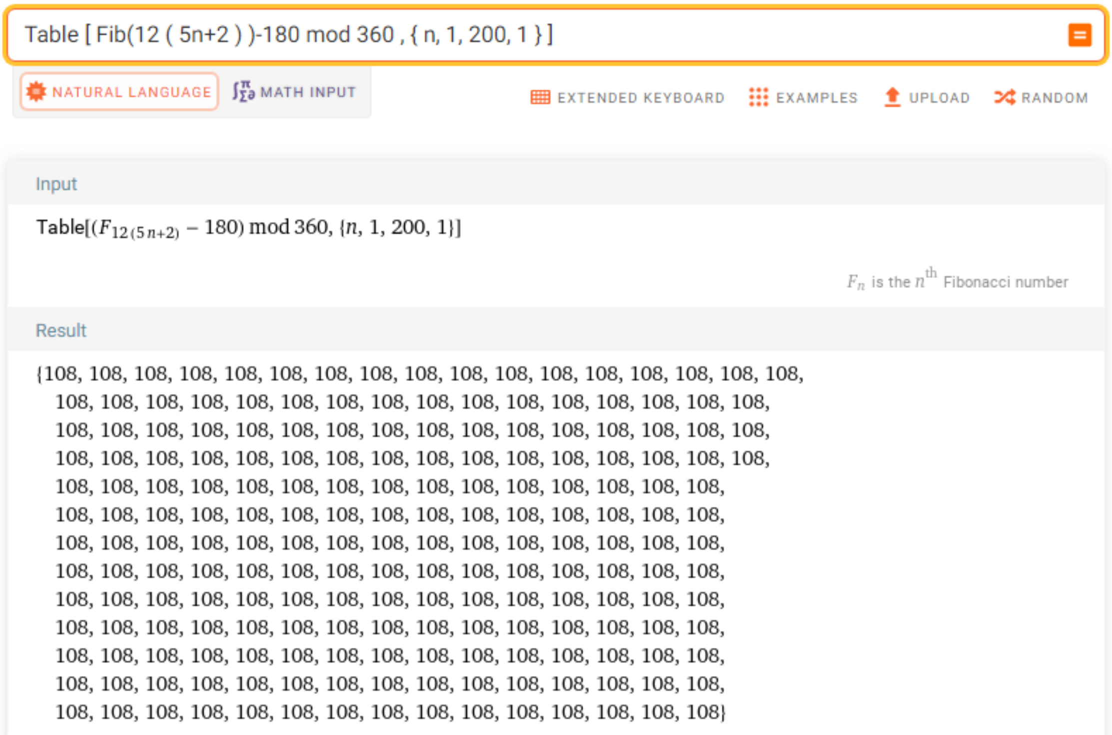

Figure 47: Table of class $F_{12k(8)} - 180$.

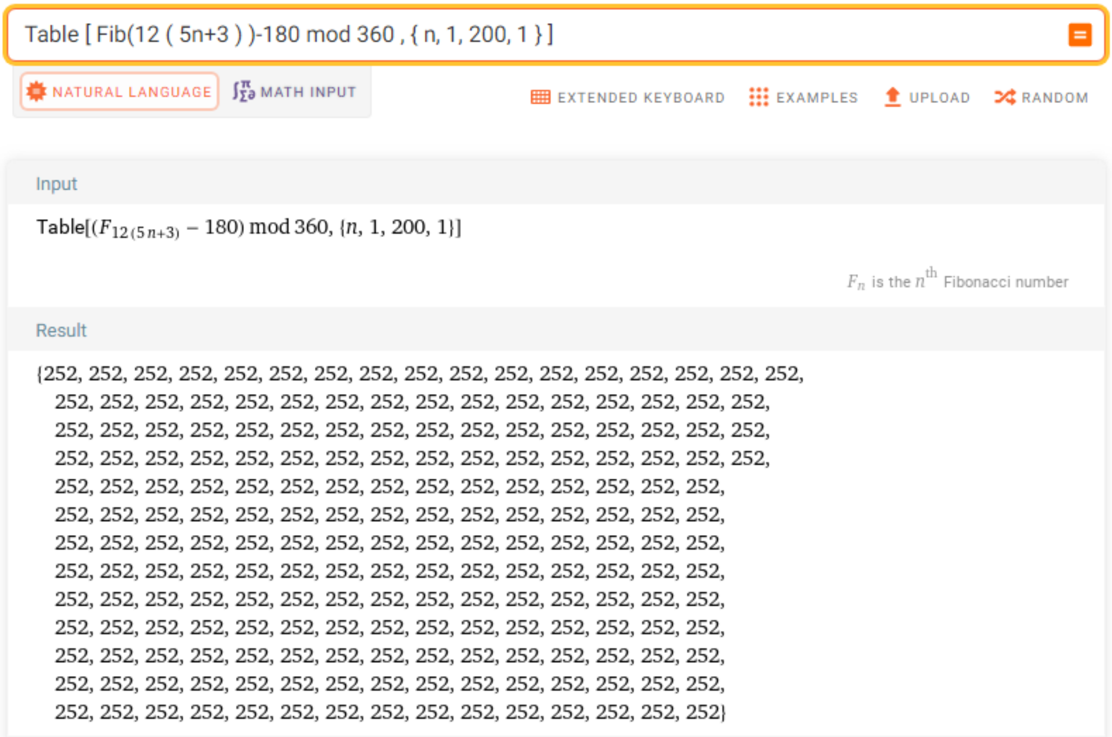

Figure 48: Table of class $F_{12k(2)} - 180$.

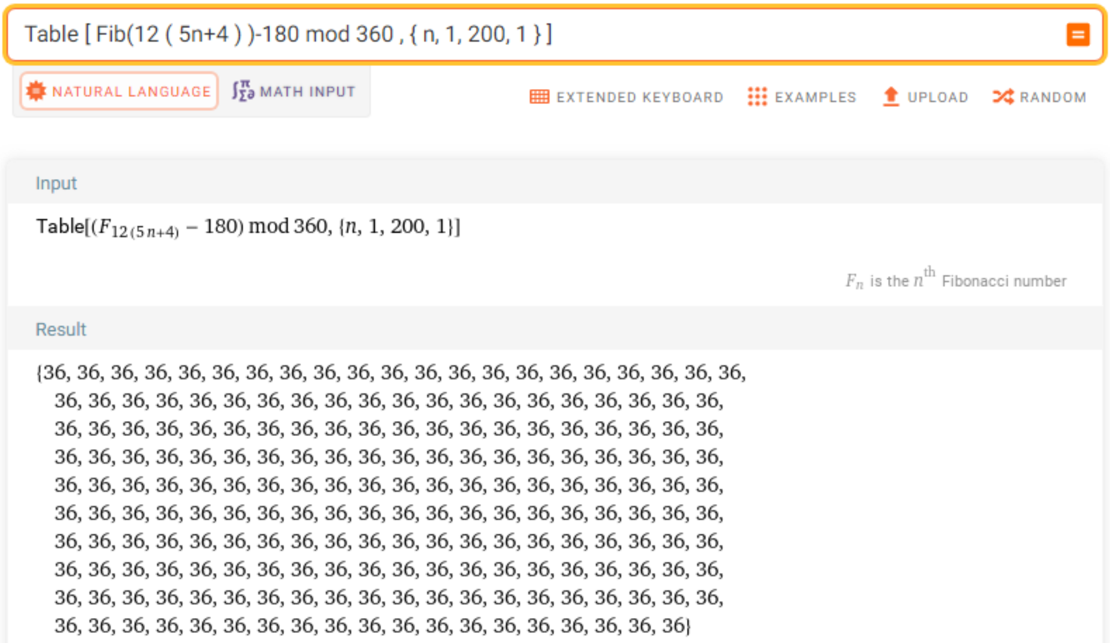

Figure 49: Table of class $F_{12k(6)} - 180$.

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9 Theorem : Golden Fibonacci Rotation Matrix

The residual classes modulo 360 of the twelfth Fibonacci correspond to isometries of regular pentagons, i.e., angles invariant under rotations that determine a rotational Lie group. Recall that an orthogonal matrix is a square matrix whose inverse matrix is equal to its transpose matrix. Rotational matrices are orthogonal matrices with determinant 1. The set of rotational matrices form a group called special orthogonal group $SO(n)$.

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \qquad det(A) = \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = 1$$

The angles $\theta = \{36°, 72°, 108°, 144°, 216°, 252°, 288°, 324°\}$ are the 8 invariant remainder classes of the twelfth Fibonacci numbers

9.1 Eigenvalues of the Golden Fibonacci Matrix

- **For classes $F_{12k(2)}$ y $F_{12k(8)}$**

  Characteristic polynomial: $\lambda^2 - \varphi^{-1}\lambda + 1 = 0$

  $\lambda_1 = \frac{\varphi^{-1}}{2} + \frac{\sqrt{2+\varphi}}{2}i = e^{i72} \qquad \lambda_2 = \frac{\varphi^{-1}}{2} - \frac{\sqrt{2+\varphi}}{2}i = e^{i288}$

  Such that: $\lambda_1.\lambda_2 = e^{i72}.e^{i288} = 1$

- **For classes $F_{12k(4)}$ y $F_{12k(6)}$**

  Characteristic polynomial: $\lambda^2 + \varphi\lambda + 1 = 0$

  $\lambda_1 = -\frac{\varphi}{2} + \frac{\sqrt{2-\varphi^{-1}}}{2}i = e^{i144} \qquad \lambda_2 = -\frac{\varphi}{2} - \frac{\sqrt{2-\varphi^{-1}}}{2}i = e^{i216}$

  Such that: $\lambda_1.\lambda_2 = e^{i144}.e^{i216} = 1$

- **For classes $F_{12k(2)} - 180$ y $F_{12k(8)} - 180$**

  Characteristic polynomial: $\lambda^2 + \varphi^{-1}\lambda + 1 = 0$

  $\lambda_1 = -\frac{\varphi^{-1}}{2} + \frac{\sqrt{2+\varphi}}{2}i = e^{i108} \qquad \lambda_2 = -\frac{\varphi^{-1}}{2} - \frac{\sqrt{2+\varphi}}{2}i = e^{i252}$

  Such that: $\lambda_1.\lambda_2 = e^{i108}.e^{i252} = 1$

- **For classes $F_{12k(4)} - 180$ y $F_{12k(6)} - 180$**

  Characteristic polynomial: $\lambda^2 - \varphi\lambda + 1 = 0$

  $\lambda_1 = \frac{\varphi}{2} + \frac{\sqrt{2-\varphi^{-1}}}{2}i = e^{i36} \qquad \lambda_2 = \frac{\varphi}{2} - \frac{\sqrt{2-\varphi^{-1}}}{2}i = e^{i324}$

  Such that: $\lambda_1.\lambda_2 = e^{i36}.e^{i324} = 1$

Eigenvectors of the Fibonacci rotation matrix

$\nu_1 = (-i, 1) \qquad \nu_2 = (i, 1)$

Let's look at some examples:

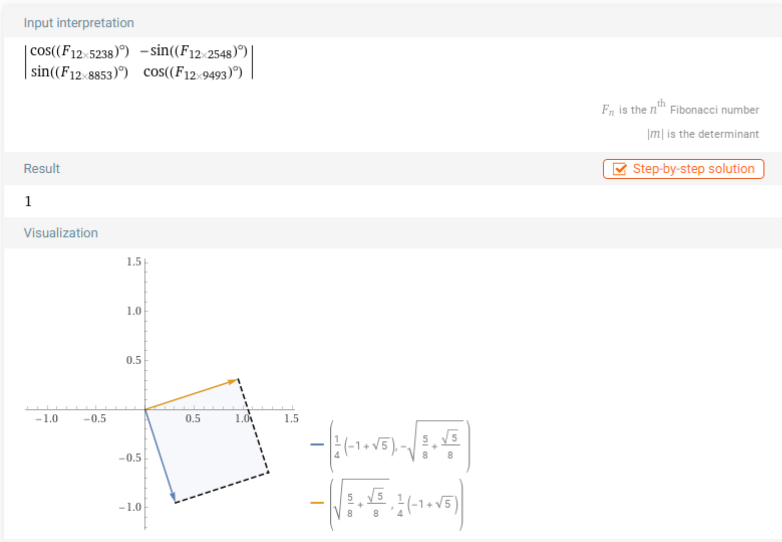

Figure 50: class $F_{12k(2)} \quad \Leftrightarrow k \equiv (3, 8) \pmod{10}$.

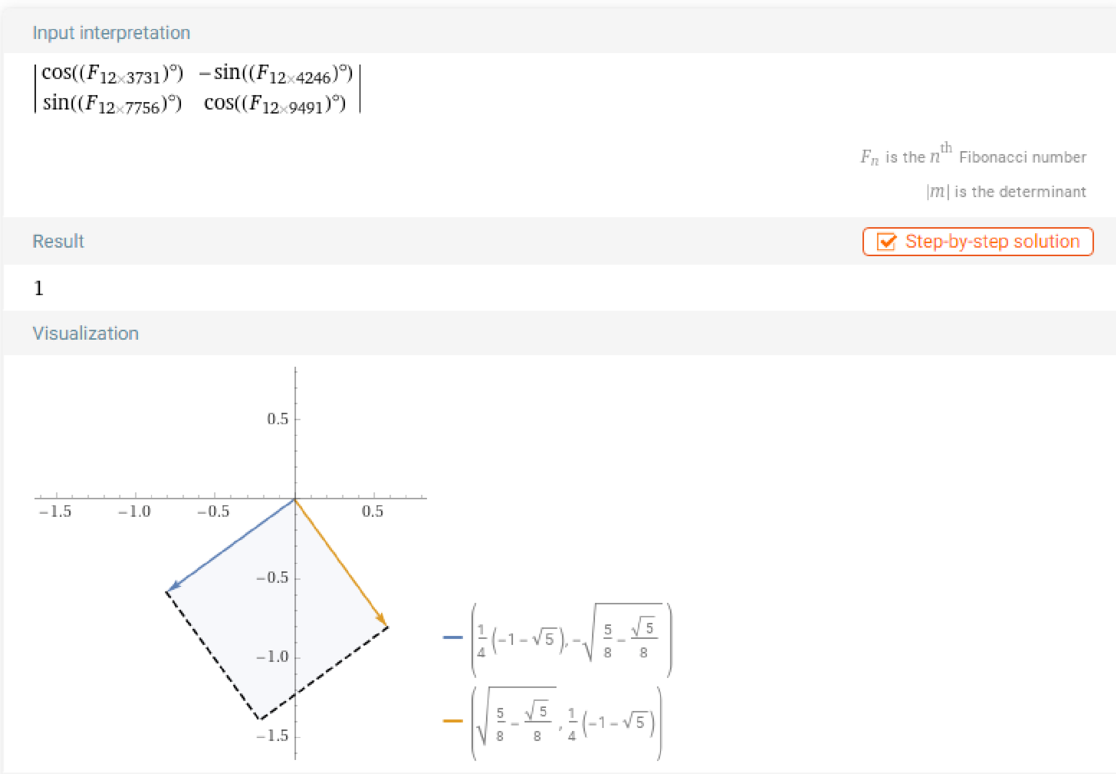

Figure 51: class $F_{12k(4)} \quad \Leftrightarrow k \equiv (1, 6) \pmod{10}$.

Input interpretation

$$\begin{vmatrix} \cos((F_{12\times 6524})^{\circ}) & -\sin((F_{12\times 78524})^{\circ}) \\ \sin((F_{12\times 4569})^{\circ}) & \cos((F_{12\times 9499})^{\circ}) \end{vmatrix}$$

F_n is the n^{th} Fibonacci number

$|m|$ is the determinant

Result

Step-by-step solution

1

Visualization

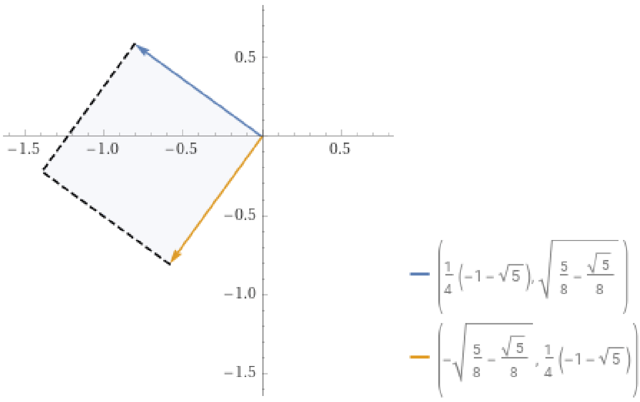

Figure 52: class $F_{12k(6)} \quad \Leftrightarrow k \equiv (4,9) \pmod{10}$.

Input interpretation

$$\begin{vmatrix} \cos((F_{12\times 6522})^{\circ}) & -\sin((F_{12\times 78527})^{\circ}) \\ \sin((F_{12\times 4562})^{\circ}) & \cos((F_{12\times 9497})^{\circ}) \end{vmatrix}$$

F_n is the n^{th} Fibonacci number

$|m|$ is the determinant

Result

Step-by-step solution

1

Visualization

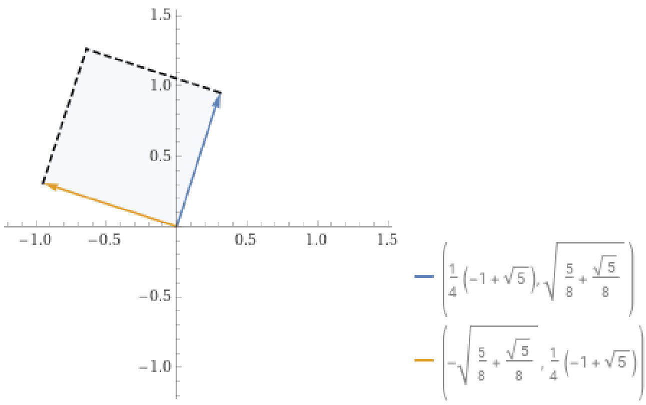

Figure 53: class $F_{12k(8)} \quad \Leftrightarrow k \equiv (2,7) \pmod{10}$.

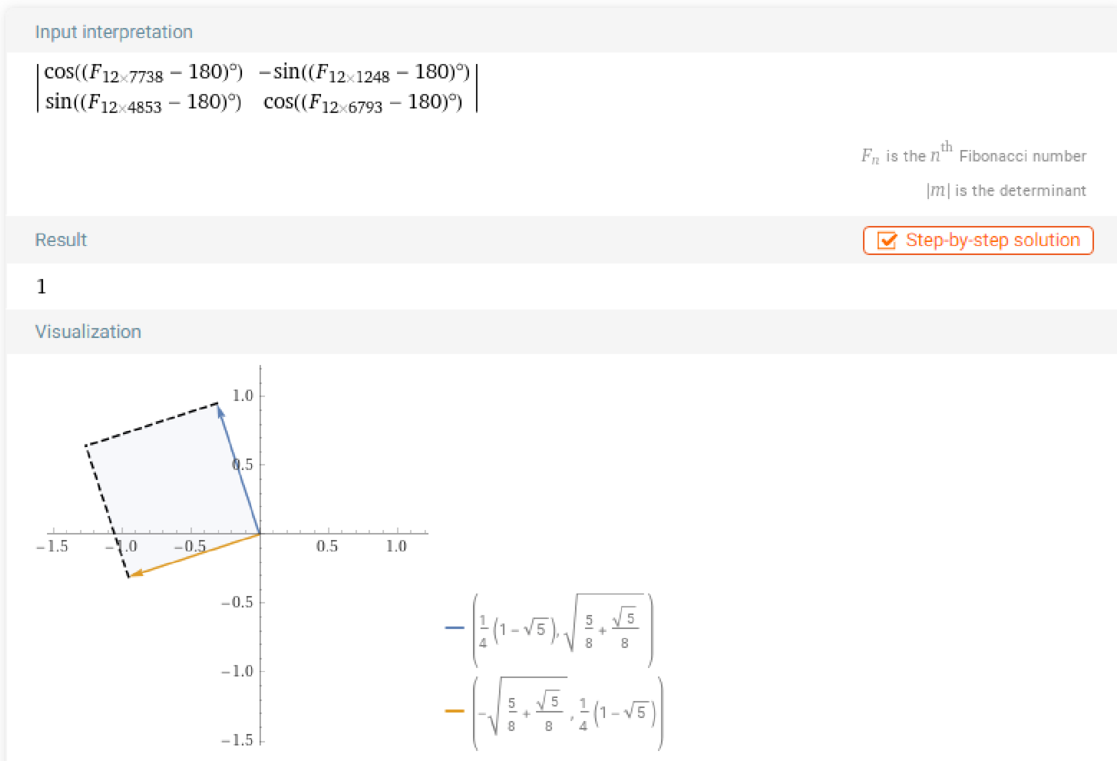

Figure 54: class $F_{12k(2)} - 180 \quad \Leftrightarrow k \equiv (3, 8) \pmod{10}$.

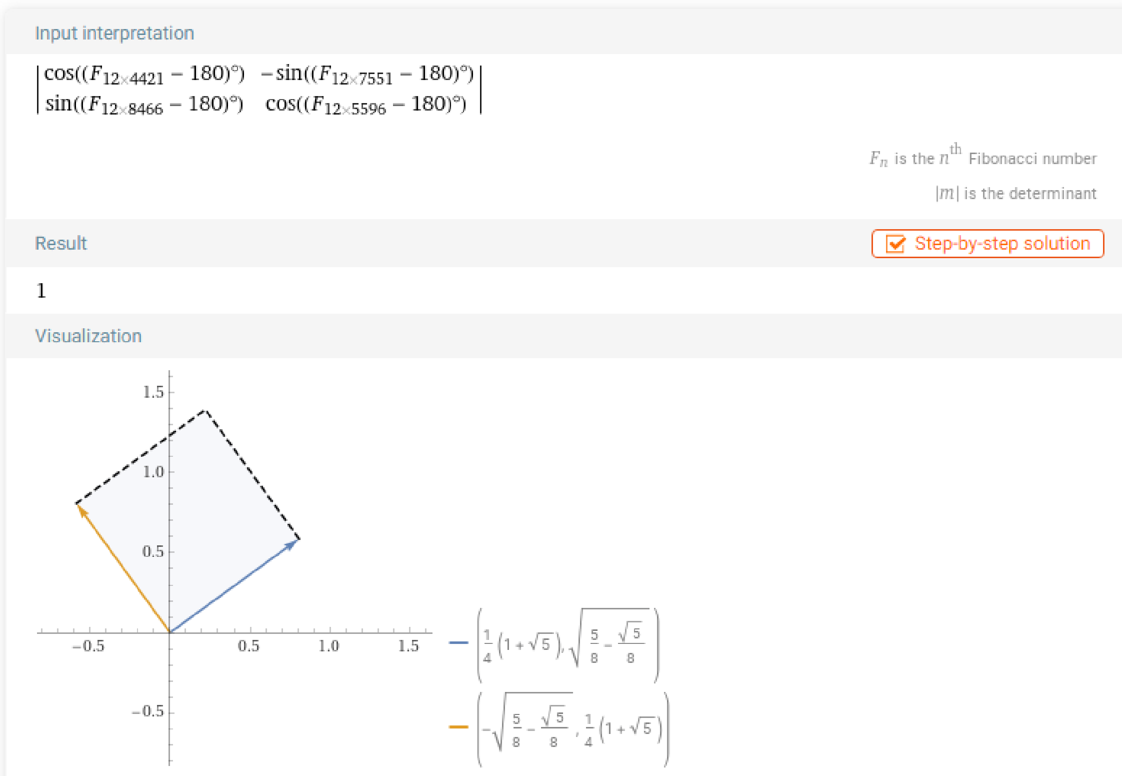

Figure 55: class $F_{12k(4)} - 180 \quad \Leftrightarrow k \equiv (1, 6) \pmod{10}$.

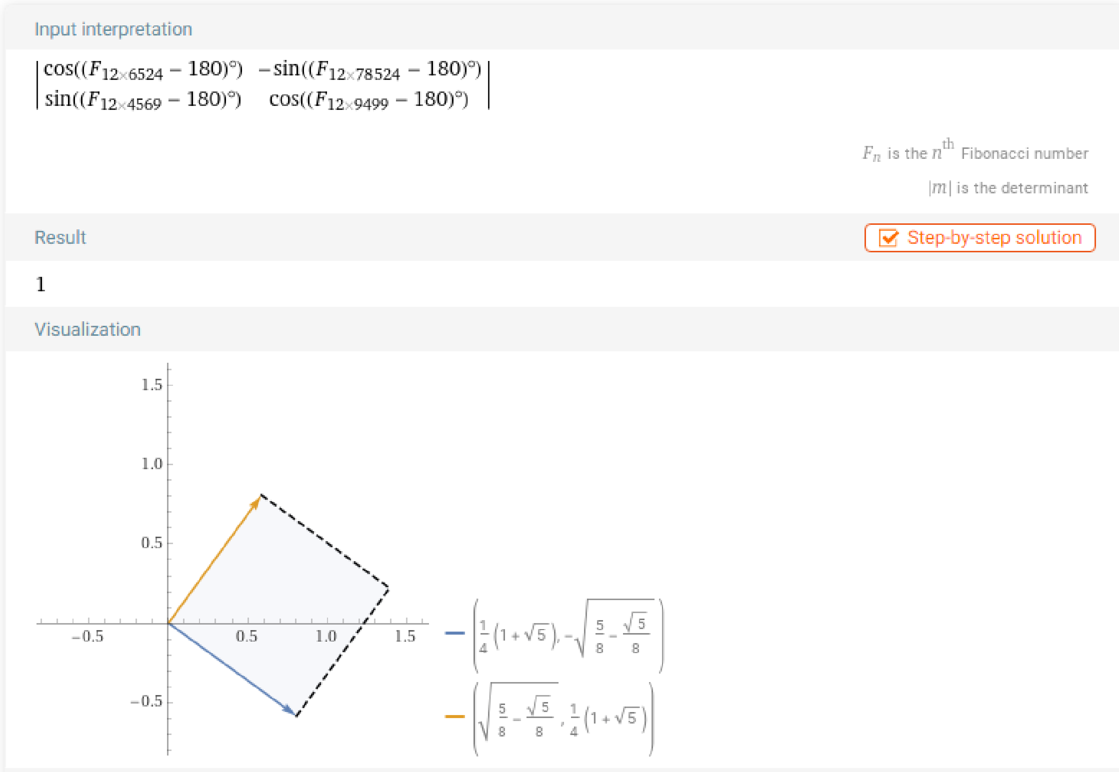

Figure 56: class $F_{12k(6)} - 180 \quad \Leftrightarrow k \equiv (4, 9) \pmod{10}$.

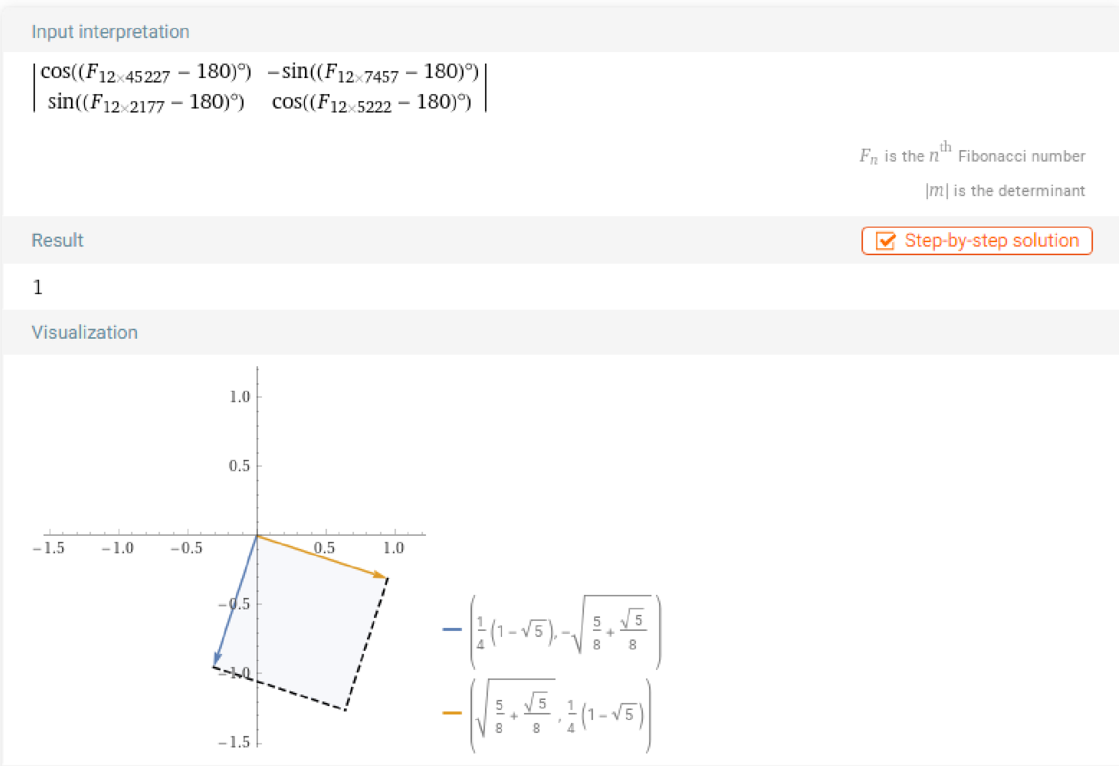

Figure 57: class $F_{12k(8)} - 180 \quad \Leftrightarrow k \equiv (2, 7) \pmod{10}$.

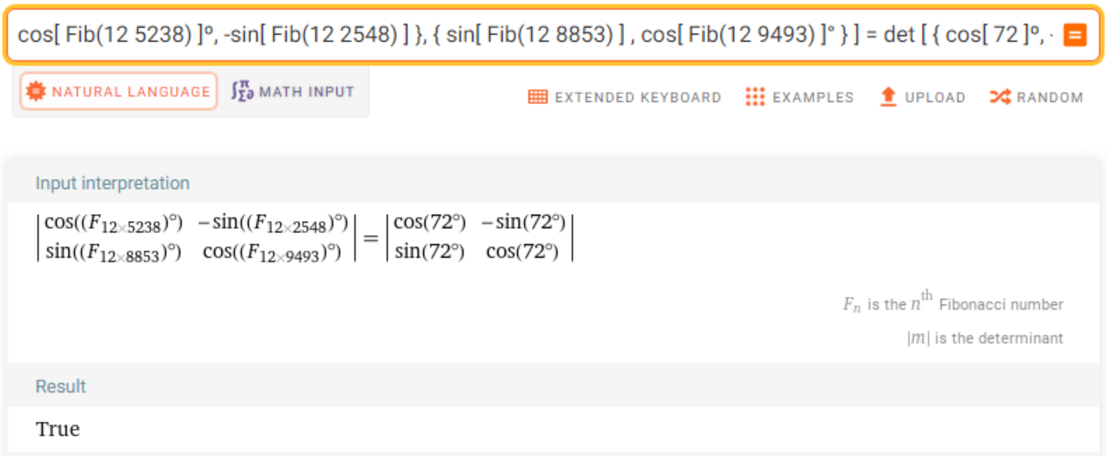

Figure 58: Class $F_{12k(2)}$.

Figure 59: Class $F_{12k(4)}$.

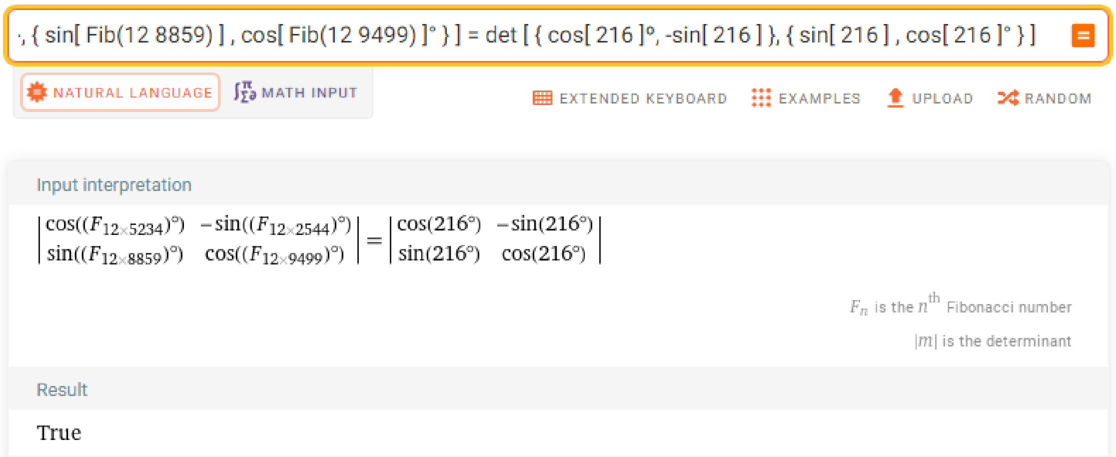

Figure 60: Class $F_{12k(6)}$.

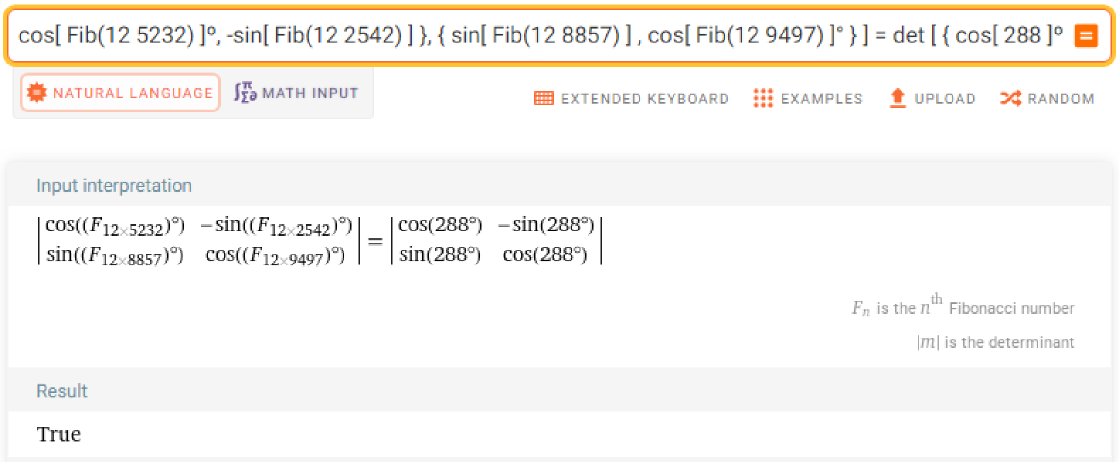

Figure 61: Class $F_{12k(8)}$.

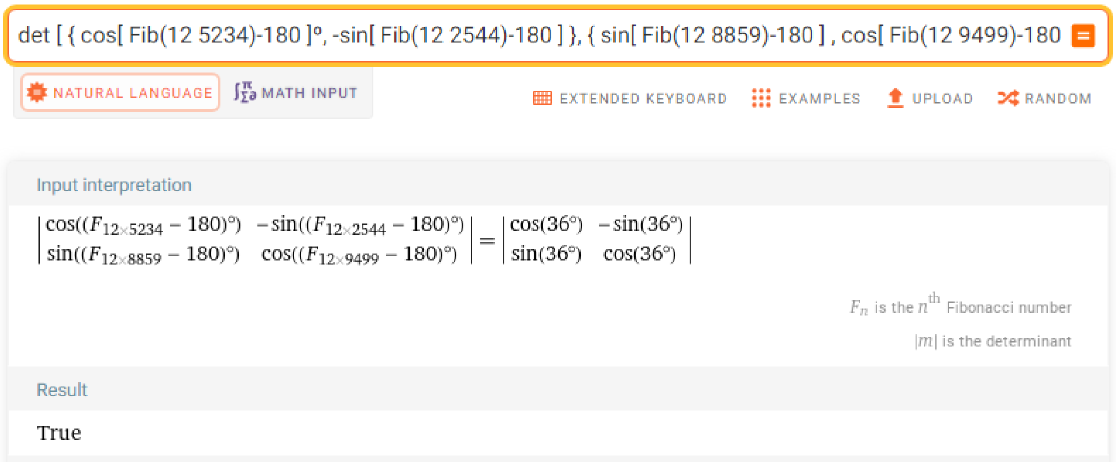

Figure 62: Table of class $F_{12k(6)} - 180$.

https://www.wolframalpha.com/input?i=det+%5B+%7B+cos%5B+Fib%2812+5233%29+%5D%C2%B0%2C+-sin%5B+Fib%2812+2543%29+%5D+%7D%2C++%7B+sin%5B+Fib%2812+8853%29+%5D+%2C+cos%5B+Fib%2812+9493%29++%5D%C2%B0+%7D+%5D+%3D+det+%5B+%7B+cos%5B+72+%5D%C2%B0%2C+-sin%5B+72+%5D+%7D%2C++%7B+sin%5B+72+%5D+%2C+cos%5B+72+%5D%C2%B0+%7D+%5D&lang=es

10 Theorem 7: Golden Fibonacci Minimal polynomial

10.1 Golden Fibonacci Isometries

$\theta = \{36^\circ, 72^\circ, 108^\circ, 144^\circ, 216^\circ, 252^\circ, 288^\circ, 324^\circ\}$

$$\theta = \{36^\circ, 72^\circ, 108^\circ, 144^\circ, -144^\circ, -108^\circ, -72^\circ, -36^\circ\}$$
$$\theta = \left\{\frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{7\pi}{5}, \frac{8\pi}{5}, \frac{9\pi}{5}\right\} = \left\{\frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \frac{-4\pi}{5}, \frac{-3\pi}{5}, \frac{-2\pi}{5}, \frac{-\pi}{5}\right\}$$

In the complex plane we have:

$$i^{2/5} = (-1)^{1/5} = e^{\pi i/5} = e^{i36^\circ} = \cos(36)^\circ + i\sin(36)^\circ$$
$$i^{4/5} = (-1)^{2/5} = e^{2\pi i/5} = e^{i72^\circ} = \cos(72)^\circ + i\sin(72)^\circ$$
$$i^{6/5} = (-1)^{3/5} = e^{3\pi i/5} = e^{i108^\circ} = \cos(108)^\circ + i\sin(108)^\circ$$
$$i^{8/5} = (-1)^{4/5} = e^{4\pi i/5} = e^{i144^\circ} = \cos(144)^\circ + i\sin(144)^\circ$$

$$i^{-2/5} = -i^{8/5} = -(-1)^{4/5} = e^{-\pi i/5} = e^{i324^\circ} = \cos(324)^\circ + i\sin(324)^\circ$$
$$i^{-4/5} = -i^{6/5} = -(-1)^{3/5} = e^{-2\pi i/5} = e^{i288^\circ} = \cos(288)^\circ + i\sin(288)^\circ$$
$$i^{-6/5} = -i^{4/5} = -(-1)^{2/5} = e^{-3\pi i/5} = e^{i252^\circ} = \cos(252)^\circ + i\sin(252)^\circ$$
$$i^{-8/5} = -i^{2/5} = -(-1)^{1/5} = e^{-4\pi i/5} = e^{i216^\circ} = \cos(216)^\circ + i\sin(216)^\circ$$

Rotational isometries taken to the complex plane translate into fifth roots.

$$i^{4/5} = e^{i72^\circ} = \cos(F_{12k(2)})^\circ + i\sin(F_{12k(2)})^\circ$$
$$i^{8/5} = e^{i144^\circ} = \cos(F_{12k(4)})^\circ + i\sin(F_{12k(4)})^\circ$$
$$i^{-4/5} = e^{i288^\circ} = \cos(F_{12k(8)})^\circ + i\sin(F_{12k(8)})^\circ$$
$$i^{-8/5} = e^{i216^\circ} = \cos(F_{12k(6)})^\circ + i\sin(F_{12k(6)})^\circ$$

Now the rotated pentagon 180°

$$i^{2/5} = e^{i36^\circ} = \cos(F_{12k(6)} - 180)^\circ + i\sin(F_{12k(6)} - 180)^\circ$$
$$i^{6/5} = e^{i108^\circ} = \cos(F_{12k(8)} - 180)^\circ + i\sin(F_{12k(8)} - 180)^\circ$$
$$i^{-2/5} = e^{i324^\circ} = \cos(F_{12k(4)} - 180)^\circ + i\sin(F_{12k(4)} - 180)^\circ$$
$$i^{-6/5} = e^{i252^\circ} = \cos(F_{12k(2)} - 180)^\circ + i\sin(F_{12k(2)} - 180)^\circ$$

The angles $\theta = \{36^\circ, 72^\circ, 108^\circ, 144^\circ, 216^\circ, 252^\circ, 288^\circ, 324^\circ\}$ con $\varphi = \dfrac{1+\sqrt{5}}{2}$

$$z_1 = e^{i36} = \cos(36)^\circ + i\sin(36)^\circ = \frac{\varphi}{2} + \frac{\sqrt{2-\varphi^{-1}}}{2}i$$
$$z_2 = e^{i72} = \cos(72)^\circ + i\sin(72)^\circ = \frac{\varphi^{-1}}{2} + \frac{\sqrt{2+\varphi}}{2}i$$
$$z_3 = e^{i108} = \cos(108)^\circ + i\sin(108)^\circ = -\frac{\varphi^{-1}}{2} + \frac{\sqrt{2+\varphi}}{2}i$$
$$z_4 = e^{i144} = \cos(144)^\circ + i\sin(144)^\circ = -\frac{\varphi}{2} + \frac{\sqrt{2-\varphi^{-1}}}{2}i$$
$$z_5 = e^{i216} = \cos(216)^\circ + i\sin(216)^\circ = -\frac{\varphi}{2} - \frac{\sqrt{2-\varphi^{-1}}}{2}i$$
$$z_6 = e^{i252} = \cos(252)^\circ + i\sin(252)^\circ = -\frac{\varphi^{-1}}{2} - \frac{\sqrt{2+\varphi}}{2}i$$
$$z_7 = e^{i288} = \cos(288)^\circ + i\sin(288)^\circ = \frac{\varphi^{-1}}{2} - \frac{\sqrt{2+\varphi}}{2}i$$
$$z_8 = e^{i324} = \cos(324)^\circ + i\sin(324)^\circ = \frac{\varphi}{2} - \frac{\sqrt{2-\varphi^{-1}}}{2}i$$

So the product of the conjugates is 1.

$$\bigg(\cos(36)^\circ + i\sin(36)^\circ\bigg)\bigg(\cos(324)^\circ + i\sin(324)^\circ\bigg) = 1$$
$$\bigg(\cos(72)^\circ + i\sin(72)^\circ\bigg)\bigg(\cos(288)^\circ + i\sin(288)^\circ\bigg) = 1$$
$$\bigg(\cos(108)^\circ + i\sin(108)^\circ\bigg)\bigg(\cos(252)^\circ + i\sin(252)^\circ\bigg) = 1$$
$$\bigg(\cos(144)^\circ + i\sin(144)^\circ\bigg)\bigg(\cos(216)^\circ + i\sin(216)^\circ\bigg) = 1$$

Let's look at some examples with Fibonacci classes:

Figure 63: Conjugate class Fibonacci of $F_{12k(4)} - 180$ and $F_{12k(6)} - 180$.

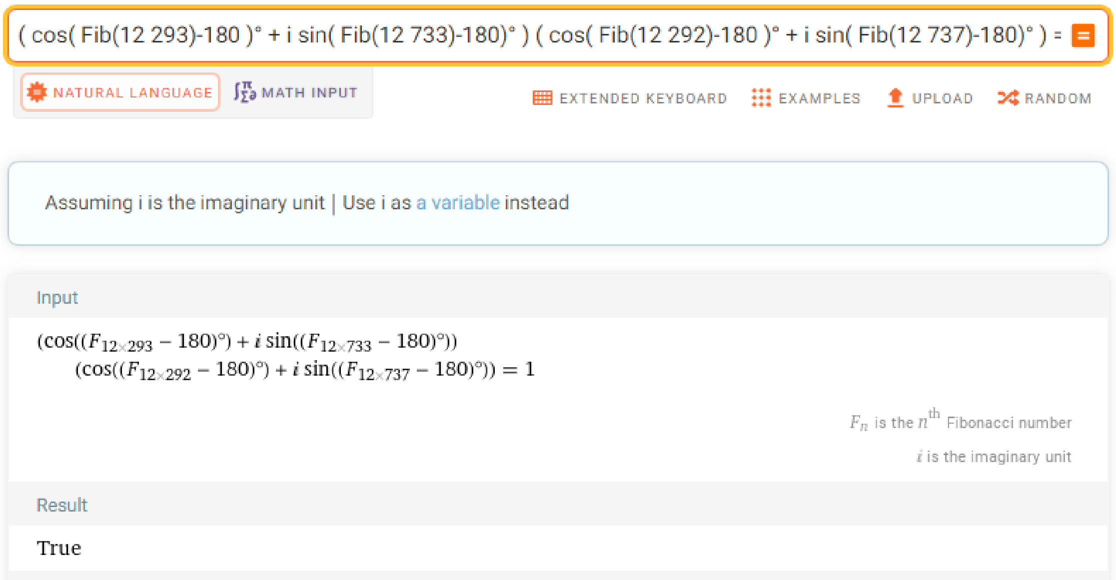

Figure 64: Conjugate class Fibonacci of $F_{12k(2)} - 180$ and $F_{12k(8)} - 180$.

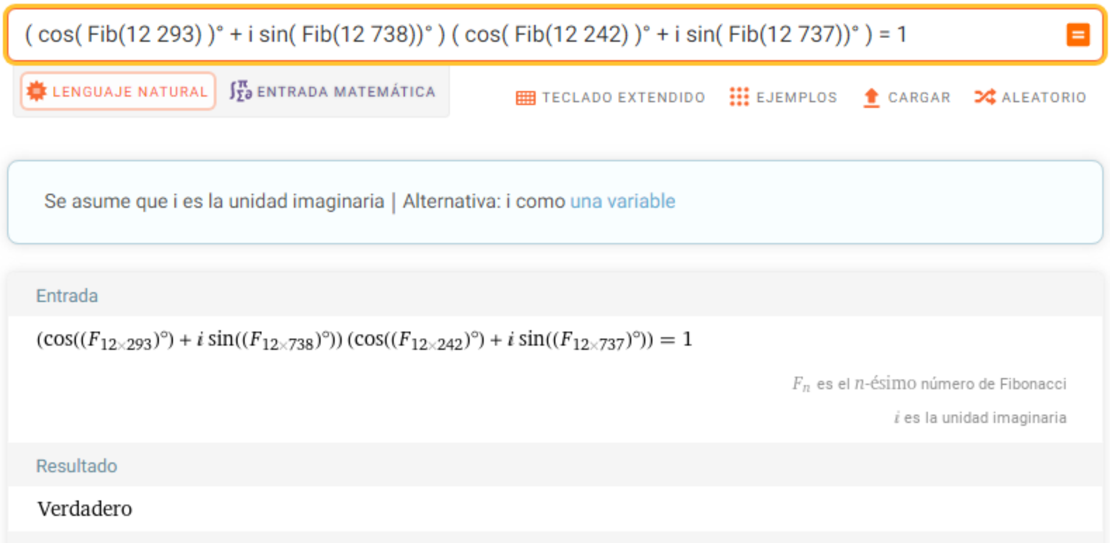

Figure 65: Conjugate class Fibonacci of $F_{12k(2)}$ and $F_{12k(8)}$.

https://www.wolframalpha.com/input?i=%28+cos%28+Fib%2812+292%29-180++%29%C2%B0+%2B+i+sin%28+Fib%2812+732%29-180%29%C2%B0+%29++%28+cos%28+Fib%2812+293%29-180++%29%C2%B0+%2B+i+sin%28+Fib%2812+738%29-180%29%C2%B0+%29++%3D+1

10.2 Golden Fibonacci Minimal Polynomial

The 8 isometries of the twelfths Fibonacci in the complex plane correspond to the zeros of the irreducible minimal polynomial over the field $\mathbb{Q}$:

$$\sum_{n=0}^{4} x^{2n} = x^8 + x^6 + x^4 + x^2 + 1 = 0$$

$$x^8 + x^6 + x^4 + x^2 + 1 = (x^4 + x^3 + x^2 + x + 1)(x^4 - x^3 + x^2 - x + 1)$$

$$= (x - i^{2/5})(x - i^{4/5})(x - i^{6/5})(x - i^{8/5})(x - i^{-2/5})(x - i^{-4/5})(x - i^{-6/5})(x - i^{-8/5})$$

$$= (x - e^{\pi i/5})(x - e^{2\pi i/5})(x - e^{3\pi i/5})(x - e^{4\pi i/5})(x - e^{-\pi i/5})(x - e^{-2\pi i/5})(x - e^{-3\pi i/5})(x - e^{-4\pi i/5})$$

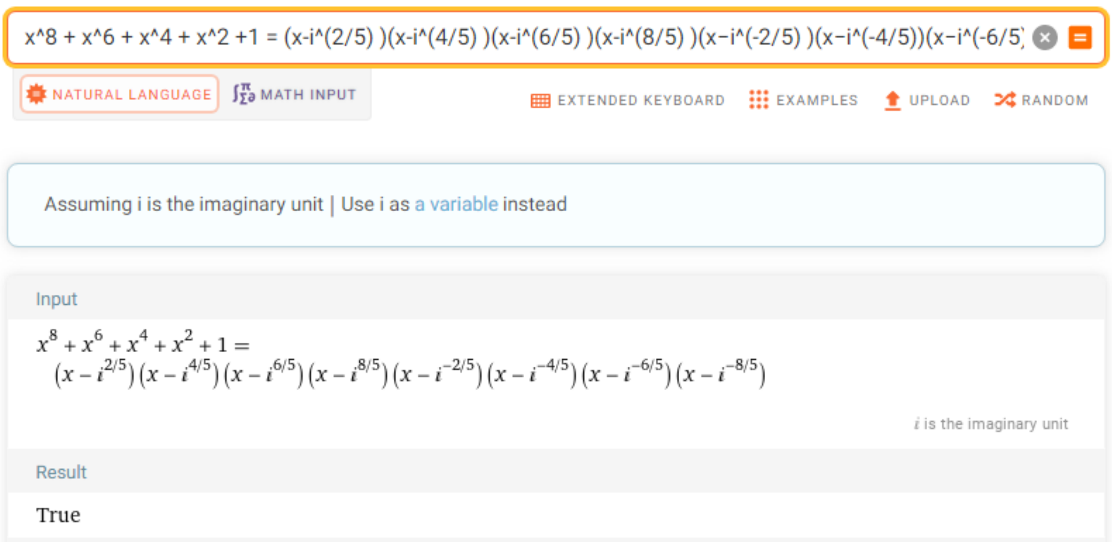

Figure 66: Zeros of minimal polynomial of Fibonacci.

https://www.wolframalpha.com/input?i=x%5E8+%2B+x%5E6+%2B+x%5E4+%2B+x%5E2+%2B1+%3D+%28x-i%5E%282%2F5%29+%29%28x-i%5E%284%2F5%29+%29%28x-i%5E%286%2F5%29+%29%28x-i%5E%288%2F5%29+%29%28x%E2%88%92i%5E%28-2%2F5%29+%29%28x%E2%88%92i%5E%28-4%2F5%29%29%28x%E2%88%92i%5E%28-6%2F5%29+%29%28x%E2%88%92i%5E%28-8%2F5%29%29

11 Theorem 8: Golden Cyclotomic Polynomials

The ancient Greek geometers posed the problem of dividing a circumference into equal parts using only ruler and compass. This problem is known today as cyclotomy. If we have a circle in the complex plane with center at the origin and radius 1, when dividing it into n equal parts it is satisfied that each of the points that cut it is a solution of the polynomial $z^n - 1$, where $z \in \mathbb{C}$, $n \in \mathbb{N}$, $n > 1$. This polynomial is defined as **cyclotomic polynomial** of order n and is denoted as Φ_n, such that it is a unit polynomial whose roots are all primitive roots of order n of unity.

Let be the roots of unity in $\mathbb{C}$ given by:

$\zeta_k := e^{2\pi ik/n}, \quad 0 \leq k < n, \quad \gcd(k,n) = 1, \quad k \in \mathbb{Z}_n$

The nth cyclotomic polynomial is defined:

$\Phi_n(x) = \prod_{k=1}^{n}(x - \zeta_k)$

The n-th roots of unity form with multiplication a cyclic group of order n, and in fact these groups comprise all the finite multiplicative subgroups of the complex numbers except the trivial $\{0\}$

The sum of the unit roots is 0, for $n > 2$. Another reason for the null sum is that the roots of unity, drawn on the complex plane, form the vertices of a regular polygon whose barycenter (by symmetry) is on the origin. This summation is a special case of the Gaussian sum.

The product of the 8 roots gives 1 and the sum gives 0.

$$(e^{\pi i/10})(e^{3\pi i/10})(e^{7\pi i/10})(e^{9\pi i/10})(e^{-\pi i/10})(e^{-3\pi i/10})(e^{-7\pi i/10})(e^{-9\pi i/10}) = 1$$

$$e^{\pi i/10} + e^{3\pi i/10} + e^{7\pi i/10} + e^{9\pi i/10} + e^{-\pi i/10} + e^{-3\pi i/10} + e^{-7\pi i/10} + e^{-9\pi i/10} = 0$$

In this work we obtained 3 irreducible minimal polynomials over $\mathbb{Q}$, whose zeros are the isometries of the 8 prime classes and the 8 classes of the twelfths Fibonacci. These irreducible minimal polynomials over $\mathbb{Q}$ are cyclotomic.

Cyclotomy of order 5

$$x^5 - 1 = (x-1)(x^4 + x^3 + x^2 + x + 1)$$
$$x^{10} - 1 = (x+1)(x-1)(x^4 + x^3 + x^2 + x + 1)(x^4 - x^3 + x^2 - x + 1)$$
$$x^{20} - 1 = (x+i)(x-i)(x^{10} - 1)(x^8 - x^6 + x^4 - x^2 + 1)$$
$$x^8 + x^6 + x^4 + x^2 + 1 = (x^4 + x^3 + x^2 + x + 1)(x^4 - x^3 + x^2 - x + 1)$$

The golden cyclotomic polynomials of order 5 are:

$$\Phi_5(x) = \frac{x^5 - 1}{x - 1} = x^4 + x^3 + x^2 + x + 1$$

$$\Phi_{10}(x) = \frac{x^{10} - 1}{(x^5 - 1)(x + 1)} = x^4 - x^3 + x^2 - x + 1$$

$$\Phi_{20}(x) = \frac{x^{20} - 1}{(x^{10} - 1)(x + i)(x - i)} = x^8 - x^6 + x^4 - x^2 + 1$$

Cyclotomic extensions

In algebraic number theory we define a cyclotomic extension of the field $\mathbb{Q}$ to any breaking field of a cyclotomic polynomial. That is, any field of the form $\mathbb{Q}(\zeta)$ where ζ is a root of unity. The cyclotomic extension is also the decomposition field of the polynomial. Therefore, it is a Galois extension. They are also abelian.

11.1 Theorem (Kronecker-Weber)

A finite Galoisian extension of $\mathbb{Q}$ of abelian Galois group is a cyclotomic extension of $\mathbb{Q}$.

This theorem implies that an algebraic number can be written as a sum of the roots of unity. Let be α an algebraic number $\alpha = \sum_{k=1}^{n} e^{2\pi i k/n}, \quad 0 \leq k < n, \quad \gcd(k,n) = 1, \quad n \in \mathbb{N}$

if and only if, the Galois group of the decomposition field of its minimal polynomial is abelian.

i.e: $\alpha = \sqrt{5} = e^{2\pi i/5} - e^{4\pi i/5} - e^{6\pi i/5} + e^{8\pi i/5}$

i.e: $\alpha = \sqrt{5}i = e^{\pi i/10} + e^{3\pi i/10} + e^{7\pi i/10} + e^{9\pi i/10}$

Because $\sqrt{5}$ has minimal polynomial $x^2 - 5$ and its Galois group is abelian. For the cyclotomic field $K = \mathbb{Q}(\zeta_5)$ its cyclotomic extension $\mathbb{Q}(\zeta_5)/\mathbb{Q}$ is Galois and the automorphisms are given by $\zeta_5 \to \zeta_5^k$, where $k = 1, 2, 3, 4$. Galois group is cyclic isomorphic to $(\mathbb{Z}/5\mathbb{Z})^*$. Let us denote its generator as σ.

$\mathrm{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong (\mathbb{Z}/5\mathbb{Z})^* = \{1, \sigma, \sigma^2, \sigma^3\}$

The Galois theory implies that K contains a unique quadratic subfield $F = \mathbb{Q}(\zeta_5)$. All thanks to identity: $\zeta_5 - \zeta_5^2 - \zeta_5^3 + \zeta_5^4 = \sqrt{5}$.

The regulator of a quadratic field is the logarithm of its fundamental unit: $\mathbb{Q}(\sqrt{5})$ is $\log\left(\frac{1+\sqrt{5}}{2}\right)$.

12 Theorem 9: Golden Prime Cyclotomic Field

The cyclotomic field associated to the irreducible minimal golden polynomial over $\mathbb{Q}$ corresponding to the cyclotomic polynomial $\Phi_{20}(x) = x^8 - x^6 + x^4 - x^2 + 1$ is the field generated by all the roots of this polynomial. These roots are the 8 primitive roots tenths of unity, which are:

$\{e^{\pi i/10}, e^{3\pi i/10}, e^{7\pi i/10}, e^{9\pi i/10}, e^{-\pi i/10}, e^{-3\pi i/10}, e^{-7\pi i/10}, e^{-9\pi i/10}\}$

The cyclotomic field generated by these roots is denoted as $\mathbb{Q}(\zeta_{10})$, where $\mathbb{Q}(\zeta_{10})$ is a primitive root tenth of unity. In other words, it is the field of complex numbers generated by all these roots. This field is an extension of the field of rational numbers $\mathbb{Q}$ and is used to study the properties of these roots and their algebraic relations.

Golden Prime Cyclotomic Extensions

The cyclotomic field $\mathbb{Q}(\zeta_{10})$ is an extension of the field of rational numbers $\mathbb{Q}$. This field has cyclotomic subfields generated by subsets of the roots of unity. In this case, one can obtain cyclotomic extensions of $\mathbb{Q}(\zeta_{10})$ generated by the fifth and second primitive roots of unity.

$\mathbb{Q}(\zeta_{10}) = \mathbb{Q}(\zeta_2, \zeta_5)$

(i) Second-order cyclotomic extension:

Similarly, a second-order cyclotomic extension can be obtained by taking the second primitive roots of unity. The cyclotomic field generated by these roots is denoted as $\mathbb{Q}(\zeta_2)$. The second primitive roots are: $\{e^{\pi i/2} = i,\ e^{-\pi i/2} = -i\}$

(ii) Fifth-order cyclotomic extension:

A fifth order cyclotomic extension can be obtained by taking the fifth primitive roots of unity. The cyclotomic field generated by these roots is denoted as $\mathbb{Q}(\zeta_5)$. The fifth primitive roots are: $\{1, e^{2\pi i/5}, e^{4\pi i/5}, e^{6\pi i/5}, e^{8\pi i/5}\}$.

Golden Prime Galois Group

By definition: $Gal(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^*$. Then the Galois group of $\mathbb{Q}(\zeta_{10})$ over $\mathbb{Q}$ is denoted $Gal(\mathbb{Q}(\zeta_{10})/\mathbb{Q})$. Since this Galois group acts on the unit roots generating a cyclic group of order 10, one has that it is isomorphic to the unit group modulo 10, that is, $\mathbb{Z}_{10}^*$.

Therefore, $Gal(\mathbb{Q}(\zeta_{10})/\mathbb{Q}) \cong \mathbb{Z}_{10}^*$.

Characterization of $\mathbb{Z}_{10}^*$

The set of units, denoted as $\mathbb{Z}_{10}^*$, refers to the set of invertible elements in the ring $\mathbb{Z}_{10}^*$, i.e., the integers modulo 10 that are coprime with 10. The elements of $\mathbb{Z}_{10}^*$ are those integers that share no common factor with 10 except 1. Here are some properties of $\mathbb{Z}_{10}^*$:

- **Closing under multiplication:** For any $a, b \in \mathbb{Z}_{10}^*$ its product ab also belongs to $\mathbb{Z}_{10}^*$. This means that the set is closed under multiplication.

- **Neutral element:** The number 1 is the neutral element in $\mathbb{Z}_{10}^*$ in terms of multiplication. This means that for any $a \in \mathbb{Z}_{10}^*$, one has $a.\ 1 = a$.

- **Closing under potentiation:** If $a \in \mathbb{Z}_{10}^*$ y $n \in \mathbb{N}$, then $a^n \in \mathbb{Z}_{10}^*$. This follows from the property of the multiplicative inverse.

- **Multiplicative inverse:** Every element in $\mathbb{Z}_{10}^*$ has a multiplicative inverse. In other words, for each $a \in \mathbb{Z}_{10}^*$, there exists a $b \in \mathbb{Z}_{10}^*$ such that $a.b \equiv 1 \pmod{10}$. This means that every element in $\mathbb{Z}_{10}^*$ has a modularly inverse integer which, when multiplied by the original element, yields 1 when taken modulo 10. For example:

  The modular inverse of 3 is 7, since: $3.\ 7 \equiv 1 \pmod{10}$

  The modular inverse of 7 is 3, since: $7.\ 3 \equiv 1 \pmod{10}$

  The modular inverse of 9 is 9, since: $9.\ 9 \equiv 1 \pmod{10}$

  The modular inverse of 1 is 1, since: $1.\ 1 \equiv 1 \pmod{10}$

- **Cardinality:** The number of elements in $\mathbb{Z}_{10}^*$ is equal to the function $\varphi(10)$, where φ is the Euler function phi, which gives the size of the multiplicative group of integers modulo n and indicates the order of the group of units in the ring. $\mathbb{Z}/n\mathbb{Z}$. Para $n = 10$, se tiene $\varphi(10) = 4$. Therefore, $\mathbb{Z}_{10}^*$ has 4 elements.

- **Cyclicity:** $\mathbb{Z}_{10}^*$ is a cyclic group because it is generated by a single element. In this case, the generator is 3, since by raising 3 to different powers, we obtain all the elements of $\mathbb{Z}_{10}^*$:

  $3^1 \equiv 3 \pmod{10}, \quad 3^2 \equiv 9 \pmod{10}, \quad 3^3 \equiv 7 \pmod{10}, \quad 3^4 \equiv 1 \pmod{10}$

  Then, $\mathbb{Z}_{10}^*$ is a cyclic group generated by 3.

The elements of $\mathbb{Z}_{10}^{*} = \{1, 3, 7, 9\}$. These are the integers modulo 10 that do not share common factors with 10, except 1, and therefore, are invertible in $\mathbb{Z}_{10}$. Which as we can see correspond to the last digit of the primes that determine the golden prime classes.

13 Conclusions and results

The Theory of **Golden Prime Symmetry** as a new research project establishes the algebraic and geometric relation between the primality of the ring of integers $\mathbb{Z}_n$ and the golden ratio in the isometries of the regular pentagon in the complex plane. So its contributes several important unpublished theorems and findings that can be studied further from algebraic number theory, class field theory and the Langlands program. Moreover, in cryptography and cybersecurity, because it classifies and orders prime numbers by means of congruence classes in the complex plane, so it could improve the current encryption and search algorithms.

The following results were found:

- The sine (odd) function maps prime numbers into unique golden images that depend on the last digit of the prime. $(1, 3, 7, 9)$.
- The cosine (even) function maps the twelfths Fibonacci into unique golden images that depend on the last digit of the twelfths Fibonacci. $(2, 4, 6, 8)$.
- The golden ratio induces an equivalence relation in 8 residual classes modulo 360 over the infinite set of prime numbers. $p > 5$.
- The golden ratio induces an equivalence relation in 8 residual classes modulo 360 over the infinite set of the twelfths Fibonacci numbers F_{12k}, con $k > 0$.
- It was shown that the 8 golden prime classes correspond to the zeros of the cyclotomic polynomial of order 5: $\Phi_{20}(x) = x^8 - x^6 + x^4 - x^2 + 1$.
- It was shown that the 8 classes of twelfths Fibonacci correspond to the zeros of the cyclotomic polynomials of order 5:

  $\Phi_5(x) = x^4 + x^3 + x^2 + x + 1 \qquad \Phi_{10}(x) = x^4 - x^3 + x^2 - x + 1.$
- The matrices of prime golden rotation and twelfths Fibonacci preserve angles, and are therefore special orthogonal groups SO(2) y SO(3) which are Lie groups and have an associated Lie algebra.

In short, the Theory of Golden Prime Symmetry demonstrates the existence of a golden prime distribution that reveals deep hidden geometric connections between Number Theory, Group Theory, Ring and Field Theory, Lie Algebras, Real Analysis and Complex Analysis, as proposed by the Langlands program.

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Informa que:

JAVIER GRISALES
Participó con la Comunicación:
"Simetría áurea prima: Un grupo dorado de Lie"

X Encuentro Nacional de Matemáticas y Estadística
entre los días 1, 2 y 3 de junio de 2022.

Dado en Ibagué, a los 3 días del mes de junio de 2022.

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Universidad del Tolima | Facultad de Ciencias
Departamento de Matemáticas y Estadística

EL GRUPO DE MATEMÁTICAS DEL TOLIMA
(GRUPO-MAT)

Certifica que

JAVIER GRISALES HERRERA

Participó en el Seminario de Álgebra 2023A, con la charla titulada "Teoría de Simetría Áurea Prima", el día 29 de mayo de 2023.

Ibagué, Agosto 8 de 2023

Director
Depto. Matemáticas y Estadística

Jesús Antonio Avila G.
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